

Mayors strike back: Evidence from land use regulation*

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Abstract

While the decentralization literature traditionally frames policy design as a question of optimal authority assignment across governance levels, authority is often shared in practice. This paper studies strategic responses of local governments when a centrally imposed reform affects instruments they partially control. We exploit the 2014 nationwide abolition of floor area ratios (FAR) in France, a housing policy intended to expand supply and improve affordability. Using detailed data on zoning regulations, building permits, housing density, and prices, we show that municipalities substitute toward other regulatory instruments remaining under their control. This resulted into a slower growth in density in areas previously covered by FAR.

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1 Introduction

Authority over key policy instruments is often shared across levels of government, from environmental regulation to education and housing. Yet most economic analyses of governance begin with a binary choice between centralization and decentralization, highlighting a fundamental trade-off: centralized governments internalize cross-jurisdictional externalities, while decentralized governments tailor policies to local preferences (Tiebout 1959; Oates 1972; Alesina and Spolaore 1997; Besley and Coate 2003). In practice, however, authority rarely lies entirely on one level, as different tiers of government may retain authority over complementary policy dimensions. Even in decentralized policies, central governments often retain influence by defining the set of policy instruments available to local governments (Blanchard and Shleifer 2001). When central and local governments pursue different objectives, shared authority creates scope for vertical strategic interaction, shifting the question from who governs to how governments respond to one another.

Housing policy provides a particularly sharp setting for this question. Restrictive land use regulations are widely seen as a major driver of housing unaffordability in productive cities, with substantial costs for labor mobility and aggregate productivity (Duranton and Puga 2023). Central governments, which are more likely to internalize these aggregate costs, have therefore sought to relax local building constraints. Local governments, by contrast, are accountable primarily to incumbent residents, who may oppose new development because of congestion, fiscal pressures, or changes to neighborhood character, rather than to the broader population that would benefit from expanded supply (Fischel 2000; Mast 2024). As a result, municipalities may use other policy instruments under their control to offset national reforms to expand supply.

This paper studies how local governments respond to a centrally imposed deregulation. We exploit a 2014 nationwide French reform that abolished maximum floor area ratios (FAR), a widely used density restriction, with the explicit aim of stimulating housing construction.

France provides an ideal setting to study policy implementation under shared authority: in its highly fragmented system of local government, small municipalities retain control over key building regulatory instruments, giving mayors both the discretion and the incentive to align implementation with local voter preferences.

To trace the regulatory responses, we digitize and assemble a novel geospatial panel of land use regulation for 118 municipalities of the Paris urban area. Using administrative planning documents, we harmonize zoning maps and track changes between 2007 and 2021 in three key regulatory instruments: floor area ratios, maximum building heights, and building coverage ratios (BCR).¹

Our panel data deliver several new stylized facts on how French municipalities regulate density. First, municipalities rely on a bundle of partially redundant instruments, rather than on a single one. In 2014, FAR covered 52% of the sample area, BCR 79%, and maximum heights 96%, and identical building envelopes can be obtained through different combinations of these three rules. Second, the intensity of regulation varies substantially not only across municipalities, but also within them: FAR coverage differs markedly across zones of the same municipality, whereas maximum heights apply almost everywhere. This within-municipality variation is systematically related to local characteristics: zones subject to FAR are initially less dense, wealthier, and have higher homeownership rates, and they also tend to face stricter and more widespread regulatory tools. Third, building permit data show that the FAR was effectively binding before the reform. The distribution of observed FARs in accepted permits exhibits a large mass point just below the regulatory limits until 2013, which disappears immediately after.

We ask whether local elected officials reacted to the removal of this regulatory tool by tightening other regulatory margins to maintain constrained building envelopes. Our identification strategy exploits the nationwide but selective impact of the reform: it applied uniformly across France while affecting only areas previously constrained by FAR limits. We implement a difference-in-differences (DID) design comparing treated and untreated areas within the Greater Paris region to estimate the causal effect of the reform on regulatory choices and housing market outcomes.

Our first finding is that municipalities offset the reform by tightening the two complementary instruments remaining under their control. In treated areas relative to control areas, maximum building heights decreased by 0.5 to 1.2 meters, corresponding to 3 to 7% of the pre-reform mean, while building coverage ratios decreased by 1 to 3 percentage points, corresponding to 2 to 6% of the pre-reform mean. We then turn to permitting

1. The building coverage ratio is the percentage of a land parcel's area that is covered by the building's footprint. By definition, it is between 0 (no building) and 1 (the building covers the full plot, with no open spaces). See Figure A.2.

decisions. Using an exhaustive administrative dataset covering all building permit applications, we show that the probability an application is denied increases by 2 to 5 percentage points, corresponding to 12 to 29% of the pre-reform mean. These results are robust to exploiting within-municipality variation accounting for municipality-level shocks.

We then examine housing market outcomes using comprehensive administrative data on the housing stock and real estate transactions. We find that density growth was significantly slower in treated areas, with the gap widening over the sample period. We find no detectable effect on prices — a null that is itself consistent with the small price response implied by standard demand elasticities given the modest change in effective density. These findings suggest that local regulatory adjustments substantially dampened the reform’s intended effect on housing supply.

Next, we ask whether the strategic interaction between tiers of government may be shaped by party affiliation. Indeed, local elected officials who belong to the government party may share the government’s policy goals, or can be inclined to implement its platform locally due to party discipline or simply better coordination.² We thus ask whether politically aligned municipalities reacted differently to the reform compared to others. We find no evidence that differences in partisanship and political alignment matter, as both right- and left-wing municipalities adjusted their land use regulation. Possibly, local elected officials may counteract the policies decided by the government — even when they belong to the government party — if their constituents oppose densification. To test this hypothesis, we consider the win margin of the mayor and compare the municipalities where the mayor was elected with a comfortable margin to others where it was narrower. We find that building constraints were specifically tightened in municipalities where the mayor’s seat is less safe. Taken together, these results suggest that local regulatory tightening is primarily driven by local electoral incentives, rather than by shared policy objectives or coordination capacity.

We then ask which local interests drive the regulatory backlash. A natural candidate is the homevoter mechanism: homeowners may oppose densification because new development can reduce neighborhood amenities and threaten the value of existing housing

2. Spenkuch, Teso, and Xu 2023 for instance, show that political alignment within organizations could increase productivity.

assets (Fischel 2001). We test this channel by interacting the treatment with the municipal homeownership rate, but find no stronger tightening response where homeowners are more numerous. This suggests that homeownership is, at best, an imperfect proxy for the stakes created by densification. By contrast, we find substantially stronger tightening in municipalities with higher housing prices. This result is informative because high prices both raise the potential gains from additional floor space and increase the value of the local housing environment that residents may seek to protect. The fact that regulatory tightening is concentrated in high-price municipalities therefore suggests that the perceived costs of densification, through reduced amenities or changes in neighborhood character, dominate potential development gains. Overall, the evidence points less to ownership status per se than to the value of local housing assets exposed to densification.

These findings speak to the design of top-down housing reforms, which have become increasingly prominent tools to address housing shortages. Recent examples include Massachusetts' Section 40B, which allows developers to bypass local zoning in municipalities with insufficient affordable housing (Blanco and Sportliche 2024), and California's Housing Accountability Act, which limits municipalities' discretion to deny compliant housing applications. Our findings suggest that, when multiple tiers of government share regulatory authority, mandates from higher levels may fail to achieve their intended objectives if local governments can use their remaining discretion to offset them. This implies that effective top-down reform depends not only on changing formal rules, but also on constraining the local margins through which these changes can be undone. Where local implementation persistently diverges from central objectives, achieving reform may therefore require either limiting local discretion more directly or reallocating regulatory authority toward higher levels of government.

Our paper contributes to the political economy literature on optimal governance structures. First, it complements the literature on decentralization, which studies the allocation of authority across levels of government (Tiebout 1959; Oates 1972; Besley and Coate 2003; Enikolopov and Zhuravskaya 2007; Narasimhan and Weaver 2024). In housing, this perspective has motivated work on how jurisdictional fragmentation shapes land use regulation and housing supply (Ellickson 1971; Tricaud 2025). We shift attention from the assignment of authority across tiers (Strumpf and Oberholzer-Gee 2002) to the strategic interaction that arises when authority is shared. Second, it contributes to the literature

on strategic interactions across jurisdictions. While economists have extensively studied horizontal competition among governments at the same level, especially in taxation and public spending (Epple and Zelenitz 1981; Bewley 1981; Epple and Sieg 1999; Brueckner and Saavedra 2001; Janeba and Osterloh 2013; Agrawal 2015), much less is known about strategic interactions between central and local governments. Existing literature has primarily documented the fiscal response of lower tiers to intergovernmental grants (Baicker and Gordon 2006; Dahlberg et al. 2008; Bracco et al. 2015) or the importance of local officials' incentives (He, Wang, and Zhang 2020). We demonstrate that these dynamics extend beyond fiscal transfers. Our paper provides new causal evidence on how municipalities respond to centrally imposed national reforms, showing that when policy authority is shared, observed outcomes reflect vertical strategic interactions rather than the unilateral consequences of assigning authority to a single tier. Our findings underscore the inherent difficulty of reconciling central policy objectives with the divergent electoral incentives of local governments. Ultimately, this friction suggests that in the presence of persistent misalignment, the shared authority structure may prove unstable, creating a systemic pressure toward the upward transfer of regulatory prerogatives to higher tiers of government.

We also contribute to the literature on land use regulation. First, we provide causal evidence on how municipalities respond to a centrally imposed deregulation, complementing recent work on upzoning policies that mostly studies settings with limited vertical conflict because zoning changes are decided locally (Greenaway-McGrevy and Phillips 2023; Anagol, Ferreira, and Rexer 2023; Nagpal and Gandhi 2024; Blanco and Sportliche 2024). While this literature shows that upzoning can raise supply and lower prices where it is locally adopted, we help explain why such policies may fail when scaled up through central mandates, and point to the local political conditions under which they are accepted or resisted. Second, our results support political economy accounts of restrictive land use regulation rooted in local electoral accountability, although we qualify the specific role of homeowners in shaping local land use regulation (Fischel 2001; Hilber and Robert-Nicoud 2013; Mast 2024). Third, we build a comprehensive geospatial panel of zoning rules over fifteen years, allowing us to track regulatory changes across multiple margins. This extends recent efforts to measure land use regulation using administrative sources (Asahi, Herrera, and Silva 2022; Rollet 2025), in contrast with work relying on

surveys, cross-sections, or inferred restrictions (Gyourko, Saiz, and Summers 2008; Jackson 2016; Büchler and Lutz 2021; Gyourko, Hartley, and Krimmel 2021; Kulka 2019; Bartik, Gupta, and Milo 2024; Cui 2022). By combining zoning rules with permit denials, we also show that tightening on one regulatory margin is accompanied by stricter behavior on another.

Our paper is structured as follows. Section 2 outlines the general framework of land use regulation in France and introduces the natural experiment we exploit. Section 3 describes the data used to assess the impact of the reform on both land use regulation and the real estate market. Section 4 presents our main empirical framework. In Section 5, we present our main results and perform robustness checks in Section 6. We explore mechanisms by investigating heterogeneous responses in Section 7. Finally, Section 8 concludes.

2 Institutional setting

2.1 Local governance of zoning in France

Municipalities are the base political and administrative jurisdiction of the French state. Following decentralization reforms in 1982 and 2003, they operate under free administration, granting them legal and budgetary independence. While the central State performs a legality check on municipal acts, no intermediate level of government has the authority to override municipal decisions. This structure is characterized by extreme territorial fragmentation, with approximately 34,000 municipalities. This granular division ensures that local authority is dispersed across a large number of jurisdictions. Within this landscape, mayors hold substantial discretionary power, particularly regarding land-use regulation.

Urban planning in France operates under a hierarchical legal system in which national laws take precedence over local regulations. This framework enforces consistency in national planning standards while allowing municipalities to adapt rules to local preferences. Within it, municipalities periodically issue local zoning plans known as *Plans Locaux d’Urbanisme* (PLU).³ These plans cover a broad set of regulatory instruments:

3. Other planning documents exist at the national level but are not part of our study, except for older forms (*Plan d’occupation des sols*, POS) that were phased out in 2017. See OECD (2017) for details.

first and foremost permitted land uses and building dimensions, but also more detailed requirements such as setbacks, parking requirements, and landscaping obligations (see Appendix Section A.2.2). We focus on the three most focal regulatory tools, i.e. floor-area ratios, building heights, and building coverage ratios. These three variables have first-order implications for the building envelope allowed on each lot.

2.2 The Greater Paris region

The Greater Paris area provides an ideal setting to study strategic interactions in the housing market between central and local governments. It exemplifies highly productive metropolitan areas, such as London, New York, or Amsterdam, that face acute housing affordability pressures. In addition, its governance is highly fragmented, with 131 municipalities each retaining their own authority over land-use regulation. This institutional setting fosters political competition across municipalities (Ellickson 1971; Tricaud 2025). It creates strong incentives for local governments to use their regulatory authority to respond to voter preferences over development. Although in recent years national legislation has introduced strong incentives to delegate planning competences to higher-level intermunicipal authorities (Tricaud 2025), evidence shows that municipalities of the Greater Paris area have been particularly reluctant to relinquish control over urban planning powers (Cocquière et al. 2022).

2.3 Reforms

The 2012 Law: A First attempt In March 2012, two months before the presidential election, the central government passed a law mandating a 30% increase in building envelope nationwide. The law faced immediate resistance from local officials, who argued it imposed a one-size-fits-all standard that ignored local market conditions. It was repealed by the incoming government in August 2012 before ever taking effect.

The 2014 ALUR Act: FAR removal The new government proposed a new reform aimed at reducing municipal discretion over land use. Announced in late 2012, debated in 2013 and enacted on March 24, 2014, the *Accès au Logement et Urbanisme Rénové* (ALUR) Act eliminated municipalities' ability to enforce maximum FARs. Local gov-

ernments could no longer deny building permits for non-compliance with this criterion.⁴ Further details about the policy shocks are provided in Appendix Section A.1.4. Importantly, this law left building coverage ratios and maximum height restrictions in mayors' hands. As illustrated in Figure A.2, different combinations of FAR, BCR, and height can generate identical building envelopes. Two lots with the same envelope may therefore differ in which regulation is actually binding: in one, FAR may be the only actually constraining regulation; in the other, height or BCR may already fully determine buildable surface, making FAR requirements not binding. Consequently, abolishing FAR without altering height or BCR constraints implies that lots with identical pre-reform zoning envelopes may experience different changes in allowable density after the reform.

Amending local zoning plans Local zoning plans can be modified through a process initiated by local elected officials and further involving experts and citizens. Municipal councilors can decide to modify the plan when they deem it necessary (e.g. to address local needs, pursue new development goals, or comply with higher-level regulations). If so, this decision is followed by public consultations involving citizen participation. Local politicians, advised by urban planners and legal experts, then review the feedback and propose final adjustments to be voted on by the council. Amendments can concern different types of changes. Adjusting specific provisions, such as regulatory values, usually takes a few months to a year. Reclassifying zones generally requires two to three years. Comprehensive revisions, involving a complete updating of the plan's objectives and content, can take 10 to 15 years.

Given the heightened salience of upzoning following the 2012 law and the new government's stated ambition to increase housing supply, local officials may have anticipated the 2014 reform and adjusted zoning rules in advance. To account for potential anticipation effects, we consider 2011 as the pre-reform reference year.

3 Data

To assess the impact of the reform, we assemble original and administrative data into four estimation samples, each tailored to a specific outcome and unit of observation. They

4. The law also removed minimum lot sizes requirements. As we show in Appendix Section A.3.1.2, minimum lot sizes were rarely used nor binding, and we further justify their exclusion from our analysis.

correspond to the four panels of Table 1: a parcel-level sample of land use regulation values (Panel B), a permit-level sample of permitting decisions (Panel C), a balanced zoning block-level panel of housing supply (Panel A), and a transaction-level sample of housing prices (Panel D). All four are built around a common core, an original geospatial panel of land use regulation that we match to stable geographic units from the French land registry. We describe this data set before turning to each estimation sample. The provenance of the underlying sources is detailed in Appendix Section A.2.

3.1 A geospatial panel of land use regulation

Our first contribution lies in the digitization and assembly of an unprecedented 15-year-long geocoded panel on land use regulations, encompassing 118 of the 131 municipalities in Greater Paris. To build the panel, we combine zoning maps (the subdivision of municipalities into zones) with written regulatory documents specifying the rules for each zone.

Regulatory content Plans follow a standardized format. Each includes a map that divides the municipality into zones designated for specific uses (urban, to-be-developed, natural, agricultural). Each zone is governed by a written document of 14 articles, setting rules on different aspects of land use. The combination of these rules defines the building envelope, the three-dimensional space within which construction is permitted on a given lot.

We build two zoning maps for early 2014 (pre-reform) and 2018, which we combine with regulatory rules coded from written documents for each year over 2007–2021. Linking zoning and regulatory data by zone name yields annual building requirements at the municipality–zone–year level. The provenance of the underlying sources and the construction steps are detailed in Appendix Section A.2.

Among these provisions, we focus on three tools: floor area ratios, building coverage ratios, and maximum heights. FAR measures density as the ratio of a building’s total floor area to its lot size: higher FARs allow denser construction. BCR measures the share of a lot covered by the building footprint: higher BCRs allow more ground coverage, while lower BCRs preserve open space. Maximum heights, expressed in meters, limit building heights (but not the number of floors). Additional constraints may also shape the building

envelope, such as offset rules or parking requirements.⁵ In Appendix Section A.2.2, we provide more details on zoning maps and rules, and Figure A.2 illustrates how the three tools jointly determine the building envelope for a given lot size.

Observation units Our first goal is to analyze how local land use regulations change around the reform. Yet zoning maps are not stable over time, since they can be redrawn during regulatory reforms. We therefore rely on a more granular and stable geographical unit, the parcel, taken from *Fichiers fonciers*, the French land registry (*cadastre*)⁶ The registry is available annually from 2009 onward, except for 2010.⁷ Working at the parcel level lets us track regulatory constraints on each parcel even when it is reassigned to a new regulatory zone.

When studying outcomes like density or counts, the parcel is not a reliable unit, because parcel boundaries do not coincide with buildings: a single building often spans several parcels but is recorded on only one. Such outcomes are therefore mismeasured at the parcel level, so we aggregate them to the zoning block instead. In this case, we use the most granular yet stable-in-time unit of observation, which we call the zoning block: it is defined as the intersection between the housing block (*section cadastrale*) and the zoning map as defined in 2014. Because both components are fixed, zoning blocks are stable over time and let us build a balanced panel while preserving granular spatial variation. We use the zoning block for outcomes that must be tracked for the same area over time or summed into counts, such as housing supply or building permits. For these outcomes, we aggregate the underlying parcel-level measures up to the zoning block and analyze them as a panel. In total, our sample comprises 1,672 regulatory zones and 3,635 housing blocks, whose intersection yields 11,030 zoning blocks. These two units, the parcel and the zoning block, organize the four estimation samples described below.

Descriptive statistics In 2014, FAR covered 52% of our sample area, BCR 79%, and maximum heights 96%. As shown in Figure 1a, the use of maximum heights and BCR remained stable over the study period, while FAR use declined by 6 percentage points

5. We abstract from these rules as they depend on (unstable) parcel shapes, which are not mappable into absolute values.

6. See Appendix Figure A.3 for an illustration of these divisions.

7. Data for 2010 is missing because the dataset was initially released biennially before becoming annual.

(10%) between 2007 and 2013, before dropping to zero with the 2014 reform.⁸

Figure 1 further shows the spatial distribution of regulatory tools as of January 1, 2014. Dark blue areas indicate where the tool is used to regulate land use, light blue indicates areas that do not use the tool, and white areas are outside our sample. This includes non-urban areas, missing data, and the Paris municipality (See Section A.2.2.2 for more details on sample construction). Figures 1b–1d reveal substantial variation in FAR use within municipalities; BCR displays similar variation with more widespread distribution, while maximum heights are nearly universal. Further details on regulatory values and their evolution are provided in Appendix Section A.3.1.

3.2 Estimation datasets

Land use regulation (parcel level) To study how municipalities adjusted the regulatory instruments left under their control, we use the full land use regulation panel. We obtain an unbalanced panel of 482,912 parcels across 118 municipalities over 2007–2021. We treat it as a repeated cross-section and weight observations by parcel size. The outcomes are the maximum height and BCR values that apply to each parcel. Further details on data sources are provided in Appendix Section A.2. Specifically, panel construction steps and attrition are detailed in Appendix Section A.2.2.2.

Housing prices (transaction level) To study price levels and variations, we use housing transactions from the DV3F database, an enriched version of the official French land transaction registry provided by CEREMA, a public agency under the Ministry of Ecological Transition, for the area covered by our land use regulation panel. We recover all house and apartment sales in our area of study between 2010, the first year of available data, and 2021. We retain only transactions corresponding to the sale of houses or apartments and analyze the log price per square meter together with hedonic controls.

8. Before the removal, many municipalities were already phasing out FAR in favor of other regulatory tools, reflecting a shift toward renewed approaches to urban planning (Dorel, Julien, and Rouillon 2013). The reform accelerated this trend by forcing municipalities that would have likely abandoned FAR in a later plan revision to do so earlier. Six municipalities fully abandoned FAR between 2007 and 2013: Aubervilliers, Bondy, Tremblay-en-France, Alfortville, Cachan, and Limeil-Brévannes. These mid-sized, densely populated suburbs have lower incomes and higher poverty rates than the wealthiest parts of Greater Paris, but also face strong demographic growth and urban transformation, likely prompting plan revisions and the abandonment of FAR.

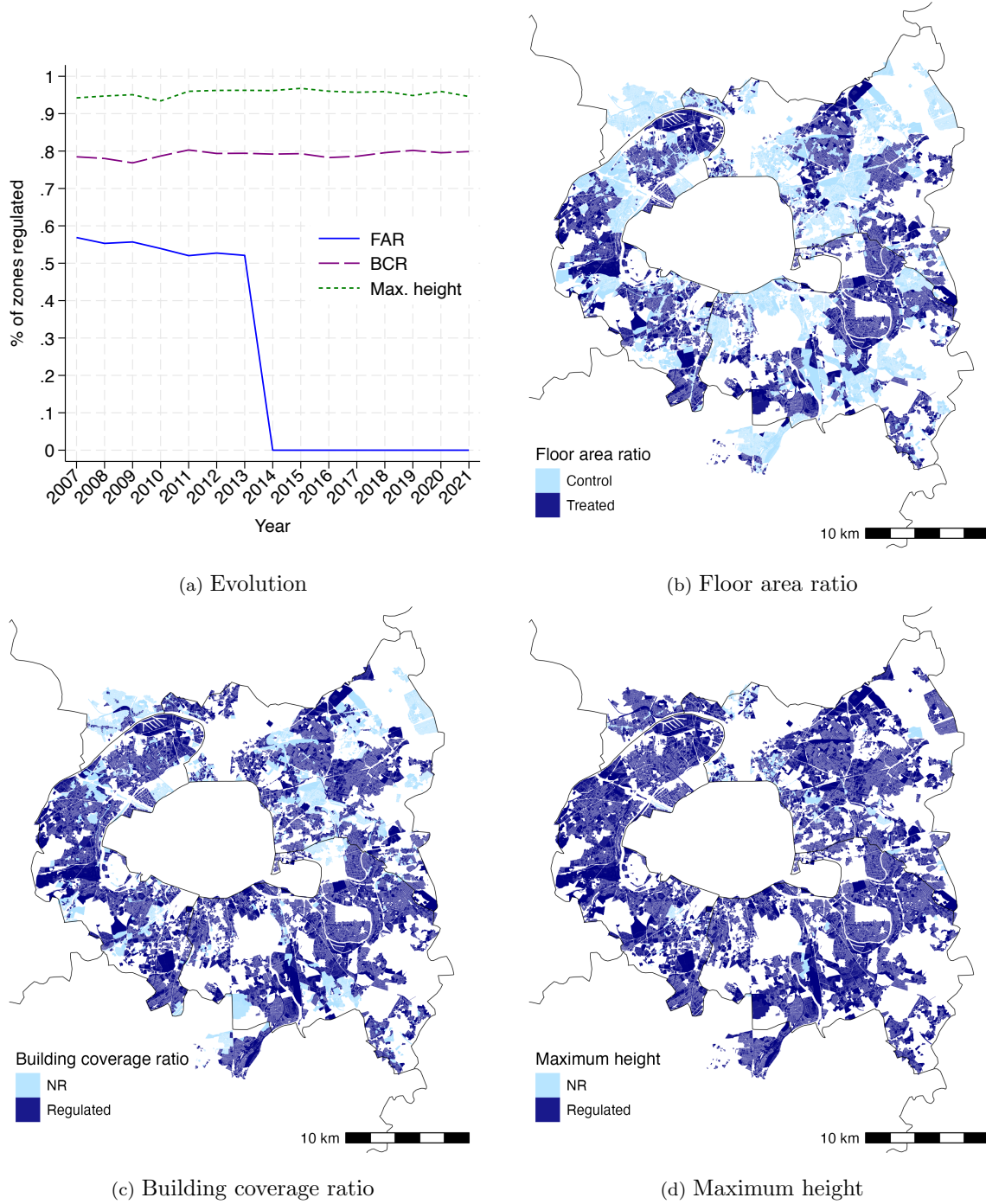


Figure 1: Areas covered by different land use measures as of January 1, 2014

Notes: The figure shows the share of zones covered by different zoning requirements over the study period, with zones weighted by their size. Figures 1b, 1c, and 1d display zones covered by FAR, BCR, and maximum height requirements, respectively, as of December 31, 2013 / January 1, 2014 in the Greater Paris area. Dark blue indicates zones subject to the corresponding regulation, while light blue indicates zones in our sample that did not use that tool. White areas are outside the scope of our study (non-urban areas, missing data, and the Paris municipality). Department boundaries are shown as thin black lines.

Permitting (permit and zoning-block level) To further investigate mayoral behavior around the reform, we use exhaustive building permit records from Sit@del2, an administrative dataset produced by the Ministry of Housing that centralizes all building permits issued in mainland France. We geolocate permits, match them to our regulatory panel and exploit the data as a repeated cross-section. For each permit, we observe the administrative decision, which we classify as either authorized or denied, the latter covering all requests that did not proceed.⁹ We also use observed floor space relative to lot size in accepted permits to assess whether FAR was binding before the reform. Second, we aggregate permit counts to the zoning-block level to study changes in the number of issued permits.

Housing supply (zoning-block level) To measure housing supply, we build a balanced panel at the zoning-block level. We use the land registry, which provides detailed information on owner characteristics, land use, occupancy status, and building features derived from fiscal data, to track the housing stock: we select all residential buildings, compute the number of dwellings and the housing surface at the parcel level, and aggregate them at the zoning-block level.

3.3 Socio-demographic covariates

We enrich all estimation samples with socioeconomic information from FILOCOM (*Fichier des Logements par Commune*), a large-scale administrative database on residential occupancy across municipalities (see Appendix Section A.2). We use these variables as baseline covariates and to construct the heterogeneity dimensions studied in Section 7.

9. The “denied” category groups three outcomes that all halt construction: an outright rejection by the mayor, a suspension of the proceeding triggered by zoning reforms, or a cancellation by the State’s local representatives during the legal conformity review. The grounds for denial overlap in practice: a mayor may block a permit either on discretionary grounds or because it fails the conformity check that mayors and the central administration conduct jointly. Mayoral discretion can therefore not be cleanly separated from compliance grounds. This distinction is not central to our analysis, however, since both channels reflect regulatory restrictiveness: stricter rules make permits less likely to comply and give mayors more scope to deny them. Suspensions and cancellations are also individually too infrequent to estimate on their own, so we pool the three outcomes into a single denial measure to gain statistical power.

4 Empirical Strategy

4.1 Baseline specification

To identify the causal impact of upzoning on regulatory behavior (land use regulations and permit issuance) and on housing market outcomes (supply and prices), we leverage pre-2014 variation in zoning rules. Prior to the reform, FAR applied to 52% of the study area and was in place in 98 of the 118 municipalities in our sample, though only in specific zones within each. This setting provides both between and within-municipality variation in exposure to the reform, which we leverage in a difference-in-differences framework.

We compare pre- and post-reform outcomes between treated and control zones, including municipality and year fixed effects. In alternative specifications, we further allow for municipality-by-year fixed effects, thereby relying solely on within-municipality variation in exposure. Our identification rests on the parallel trends assumption: absent the reform, treated and control zones would have followed similar trajectories.

Parcel- and permit-level outcomes At the parcel level, we treat the unbalanced panel as a repeated cross-section, since parcels frequently split and merge over time. We estimate the following equation:

$$Y_j = \beta \cdot FAR_{b(j)} \times 1\{t(j) > 2011\} + \alpha_{m(j)} + \delta_{t(j)} + \epsilon_j \quad (1)$$

Where Y_j is the outcome variable for parcel j , which is located in zoning block b within municipality m and observed at time t . $\alpha_{m(j)}$ are municipal fixed effects that absorb time-invariant characteristics. $\delta_{t(j)}$ are time fixed effects. $FAR_{b(j)}$ is an indicator equal to one if the zoning block b of parcel j was subject to a FAR restriction in early 2014, and $1_{t(j) > 2011}$ is an indicator for years after 2011. We set 2011 as the baseline year to account for potential anticipation in 2012, either in response to the first law (later repealed) or in expectation of the second law announced that year. In the robustness checks, we also examine each policy change as a separate event. ϵ_j is the error term, assumed to be independent across clusters and possibly correlated within clusters, with mean zero. Standard errors are clustered at the zoning block-level, which corresponds to the level of treatment variation. Observations are weighted by parcel size.

At the permit level, which also takes a repeated cross-section form, we estimate Equa-

tion 1 to assess changes in denial probabilities, interpreting the specification as a linear probability model. These regressions are unweighted.

Transaction-level outcomes For transaction-level outcomes (prices), we extend equation 1 by adding hedonic controls for housing characteristics. Controls include a single-family dummy, floor space, number of rooms, and construction vintage. Standard errors are clustered at the zoning block-level.

Zoning block-level outcomes As new construction often results from changes in the parcel layout, we study density at the zoning block-level, which provides a balanced panel. We estimate the following equation, clustering standard errors at the zoning block-level:

$$Y_{b,t} = \alpha_{m(b)} + \delta_t + \beta \cdot FAR_b \times 1\{t > 2011\} + \epsilon_{b,t} \quad (2)$$

We also aggregate to the zoning block-level when analyzing permit counts. Because 66% of observations equal zero, we adapt our empirical strategy by estimating a negative binomial model using a generalized linear specification. Let W denote the vector $(FAR_b \times 1_{t>2011}, X_{b,2011})$. We assume that, conditional on W , expected permit counts follow an exponential functional form. We estimate:

$$E[Y_{b,t} | W] = \exp \left(\alpha_{m(b)} + \delta_t + \beta \cdot FAR_b \times 1\{t > 2011\} \right) \quad (3)$$

Standard errors are also clustered at the zoning block-level.

4.2 Identification assumptions and threats

Our identification strategy relies on the assumption that, absent the reform, treated and control zones would have followed parallel trends. We assess this assumption by extending the baseline specification to flexibly account for observable heterogeneity and time-varying local shocks.

Baseline differences and selection into FAR A key concern is that pre-existing differences in zoning rules may correlate with underlying development dynamics, potentially biasing difference-in-differences estimates.

Prior to the reform, FAR and non-FAR areas differ along several observable dimensions

(Table 1). FAR zones are initially less dense, wealthier, and exhibit higher homeownership rates, and regulatory tools are somewhat more prevalent and restrictive. Such differences raise the possibility of differential development trends unrelated to the reform. We address this concern in three ways.

First, all specifications include municipality and year fixed effects, which absorb time-invariant heterogeneity across municipalities as well as common shocks affecting all areas. We also consider more granular fixed-effects specifications by replacing municipality fixed effects $\alpha_{m(j)}$ with municipality–FAR-status fixed effects $\alpha_{m(j) \times FAR_b}$. This amounts to splitting each municipality into areas that were initially subject to a FAR regulation and areas that were not, thereby controlling for unobserved heterogeneity specific to pre-reform FAR coverage within municipalities.

Second, we test for differential pre-trends using an event-study specification that interacts treatment status with year indicators:

$$Y_j = \alpha_{m(j)} + \delta_{t(j)} + \sum_{y \neq 2011} \beta_1^y \cdot FAR_{b(j)} \times 1\{t(j) = y\} + \epsilon_j \quad (4)$$

Examining the β_t^0 coefficients tests whether treated and control groups followed similar dynamics before the reform.

Third, we augment all specifications with baseline covariates interacted with time dummies. Controls include log population, log income, age of household head, housing surface per person, and share of social housing. We avoid using contemporaneous covariates, as they may be affected by treatment and introduce contamination bias. These controls capture pre-existing socioeconomic and housing conditions that may drive heterogeneous outcome dynamics, allowing us to assess whether the parallel trends assumption holds more credibly while potentially improving precision.

Heterogeneous exposure to local shocks A further concern is that FAR exposure may correlate with local shocks such as changes in local amenities or taxation affecting differentially treated and control areas. We address this possibility in two ways. First, we restrict the sample to areas located near the boundary between treated and control zones, thereby comparing observations exposed to similar local unobserved shocks. Second, we introduce jurisdiction-by-time fixed effects at both the department and municipal levels. This ensures that identification relies solely on within-jurisdiction variation, ruling out

the influence of time-varying shocks at these levels.

4.3 FAR bindingness

FAR removal can only affect housing markets if FARs constrained new construction, that is, if they were binding. We assess the bindingness of FAR in building permits. In Figure 2a, we plot the raw distribution of observed FAR in accepted building permits (built floor space divided by lot size) for the four years before the reform (2010–2013). The sample is divided between the treated group (blue) and the control group (red). The treated group exhibits pronounced bunching at FAR values of 0.4, 0.5, 0.6, and 0.7, patterns that are absent in the control group. Together, these four values account for 58% of FAR observations in regulated parcels during this period. Next, focusing on treated areas, we divide the sample into two periods: before the reform (2010–2013) and after the reform (2015–2018). In Figure 2b, we plot the deviation of the observed FAR from the regulatory FAR, using the 2014 value as the reference for post-reform years. We observe a clear bunch just below 0 prior to the reform, which disappears in the post-treatment period. This pattern provides strong evidence that FARs were binding before 2014.¹⁰

5 Main results

5.1 Land use regulation

We begin by examining the impact of the reform on land use regulations, focusing on changes in zoning plans and the probability that building permit applications were denied.

To support the parallel trend hypothesis, we estimate the event-study specification of Equation 4. Figure 3 plots yearly coefficients and their 95% confidence intervals. Figure 3a reports results for maximum heights and Figure 3b results for BCR values. Figure 3c shows evidence for permit denial probability, i.e. the intensive margin of permitting

10. Unlike FAR, we find no evidence that minimum lot size requirements constrained observed lot sizes. Minimum lot sizes were a less common tool (24% of sample area in 2011) and applied to projects on lots that were entirely undeveloped before, resulting in very few permits. Figures are presented in Figure A.8 in Section A.3.1.2 in the Appendix. The distribution of lot sizes before and after the reform is similar. Round number bunching appears in both regulated and unregulated areas, suggesting it reflects administrative features of the lot division process. In regulated areas, bunching close to regulatory limit is present both before and after the removal of minimum lot sizes, suggesting that regulatory values only followed the market. We conclude that MLS were not binding in most cases.

Table 1: Descriptive statistics (2011 data)

	(1) Without FAR	(2) With FAR
<i>Panel A.</i> Housing data (zoning block-level)		
Dwellings	193	175
Buildings	43	61
Total floor space (sq. m.)	11,525	11,798
Land area (sq. m.)	40,259	41,247
Floor space/unit (sq. m.)	65.7	72.7
Observed FAR	0.41	0.41
Population	524	453
Household head age	52.1	53.7
Housing tenure (years)	8.3	9.3
Homeowners (%)	39.0	51.0
Social housing (%)	14.8	11.3
Household income (€)	41,772	49,534
Income per person (€)	18,806	21,594
Total population	2,451,212	1,810,880
Observations	5,718	4,334
<i>Panel B.</i> Land use regulation data (parcel-level)		
1(Has BCR)	0.62	0.92
BCR	0.56	0.50
1(Has maximum height)	0.94	0.97
Maximum height (m)	15.8	14.5
1(Has minimum lot size)	0.04	0.42
Minimum lot size (sq. m.)	673	756
Observations	128,569	241,256
<i>Panel C.</i> Building permits data		
Dwellings	7.2	2.5
Lot area (sq. m.)	4225	1603
Floor space (sq. m.)	1418	550
Observed FAR	1.00	0.57
Permit denial probability	0.24	0.18
Observations	3,449	3,354
<i>Panel D.</i> Real estate transaction data		
Price	266,418	310,590
Floor space (sq. m.)	59.5	66.5
Price per sq. m.	4,565	4,681
1(Single family)	0.17	0.28
Number of rooms (€)	2.8	3.0
Observations	24,016	17,955

Notes: This table reports descriptive statistics for the reference year, 2011, comparing areas with and without a floor area ratio (FAR) in 2014, when FAR were effectively removed. Panel A uses zoning block-level data, where zoning blocks are defined as the intersection of fiscal blocks and regulatory zones. Panel B uses parcel-level data weighted by parcel size to summarize associated land use regulations. Panel C presents building permit data, while Panel D uses real estate transaction data. Columns 1 and 2 report means for areas without and with a FAR, respectively. Observed FAR is calculated as the total living area divided by land area.

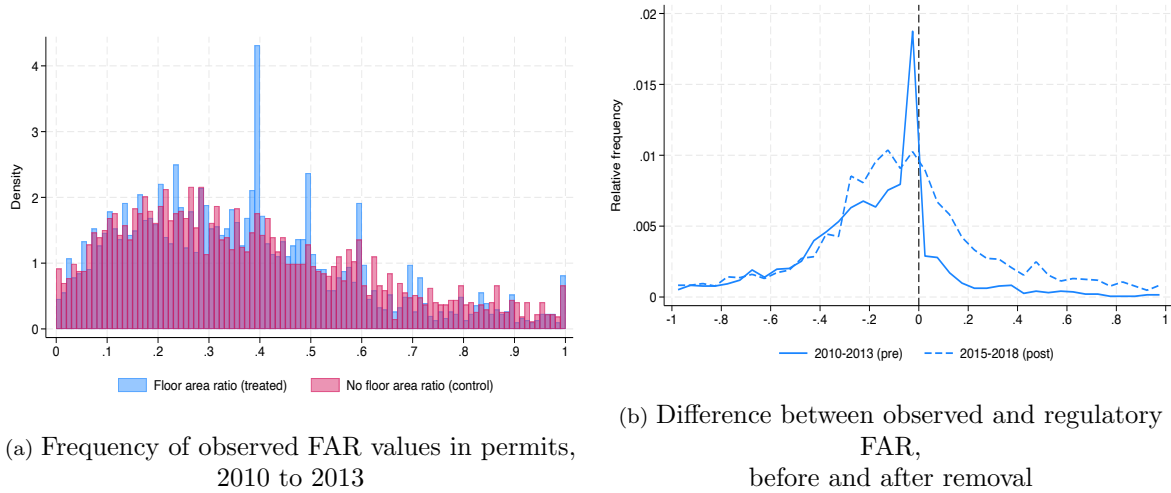
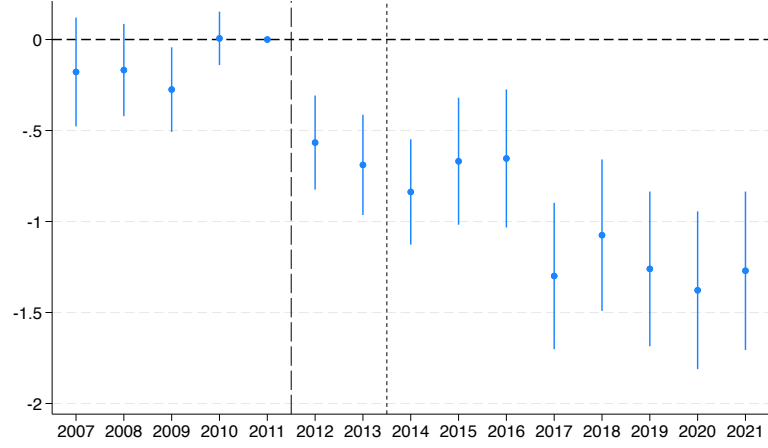


Figure 2: Observed FAR before and after the 2014 reform

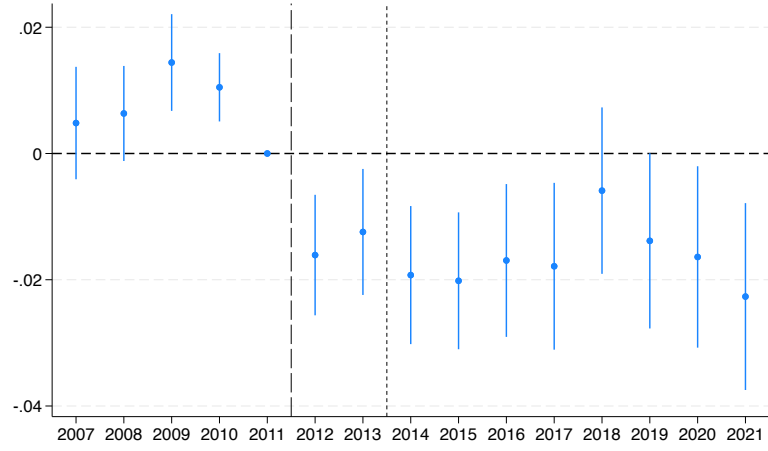
Notes: This figure shows the distribution of observed floor area ratios (FAR) for individual building permits around the 2014 reform. Observed FAR is calculated as floor space built divided by lot size. Panel a displays the raw distribution of observed FAR in the pre-reform period (2010–2013), distinguishing whether a regulatory requirement applied. Panel b plots the deviation between observed and regulatory FAR in the pre- and post-reform periods (2010–2013 and 2015–2018) in treated areas. Negative values in Panel b indicate that the project fell below the regulatory FAR, while positive values indicate that observed FAR exceeded the maximum. The latter may reflect (a) mismeasurement of lot size (in projects involving multiple parcels), (b) FAR bonuses (a 20% bonus was available under specific conditions but was almost never used, as noted by Dorel, Julien, and Rouillon 2013), or (c) permit derogations.

activity. None of the three outcomes display significant pre-trends prior to 2012, when the first reform was adopted. Table 2 presents estimates from Equation 1.

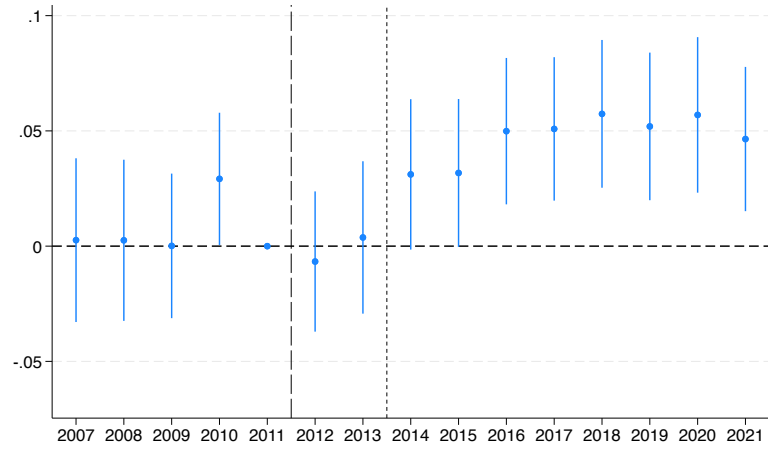
The results show that policy changes triggered downward adjustments in maximum heights and building coverage ratios in treated areas starting in 2012, suggesting that mayors responded to the reforms by tightening other regulatory tools under their control. For maximum heights, there is a sharp decline in 2012 that continues over the following years before stabilizing around 2017. Coefficients are negative and statistically significant, indicating reductions of 0.47–1.16 meters (3–7% of the 2011 mean). For BCR, Figure 3b shows a decline beginning in 2011 and intensifying after 2012. The 2011 difference is not statistically distinguishable from the 2007 baseline but lies outside the confidence interval range of subsequent years. Coefficients are negative and significant, implying reductions of 0.01–0.03 (2–6% of the 2011 mean). Finally, as shown in Figure 3c, the reform was followed by a sustained rise in permit denial probability, beginning in 2014 after the second law banning FARs took effect, peaking in 2018, and stabilizing thereafter. Relative to the pre-treatment period, denial rates increased by 2–5 percentage points (12–29% of the 2011 mean).



(a) Maximum heights value



(b) Building coverage ratio value



(c) Building permit denial probability

Figure 3: Event-study estimates of regulatory responses

Notes: Results from the estimation of Equation 4 (event study), baseline specification. Panel (a) reports coefficients when the dependent variable is the maximum height (in meters) at the parcel level. Panel (b) reports coefficients when the dependent variable is the BCR (ratio of building footprint to parcel size, ranging from 0 to 1) at the parcel level. Panel (c) reports results from a linear probability model where the dependent variable equals 1 if the building permit was denied. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. Vertical bars show 95% confidence intervals.

This evidence suggests that local authorities blocked more building permits in areas where FARs were abolished in 2014. Although urban planning documents are legally binding in France, overlapping regulatory layers create institutional complexity and ambiguity, giving local officials discretion over permit approvals. Following the removal of FARs, inconsistencies in zoning rules provided legal grounds for denial.¹¹ This mechanism explains why denial rates rise only after 2014, once FAR was effectively removed, while adjustments to maximum heights and BCR began earlier in anticipation. Tightening these regulatory tools provided a relatively low-cost and straightforward way for municipalities to protect regulated areas from the densification threat posed by upzoning reforms. Addressing inconsistencies may also require revisions to zoning documents, delaying the issuance of permits.

Additionally, many other regulatory tools remain under the authority of mayors. In particular, Articles 6, 7, and 8 of local zoning plans allow setback requirements and relative placement rules (see Section A.2.2) and command building envelope. While BCR is common, municipalities can choose not to use it and instead rely primarily on setbacks to control lot coverage and building layout. By contrast, maximum heights are defined in a single article (Article 10), which explains their near-universal use. This difference may account for why our results appear stronger for maximum heights. Since we do not observe all regulatory dimensions, our results should be seen as a conservative estimate of the regulatory reaction.

Overall, the evidence suggests that mayors reacted to the upzoning reforms both by enacting alternative forms of restrictive zoning and by denying more permits. These regulatory instruments act as substitutes for FARs, enabling municipalities to maintain density controls. Importantly, these two forms of responses appear to function as complementary tools in a broader strategy of local land use control. This interpretation is consistent with other contexts: in the UK, denial rates proxy regulatory restrictiveness in the absence of binding local plans (Hilber and Vermeulen 2016), while in the US, the Wharton Land Use Regulation Index captures approval complexity by counting the

11. In the Sitadel data, permits can be refused (explicit rejection of the application by mayors), annulled (a previously granted permit invalidated by local representatives of the central State on legal grounds), or subject to a *sursis à statuer* (a temporary suspension, used when municipalities revise their planning rules, which is rarely observed in the data). Because the same issues created by upzoning can trigger any of the three outcomes, they cannot be disentangled as separate mechanisms. Since all effectively block construction, we aggregate them into a single “denial” category to increase sample size.

agencies involved (Gyourko, Saiz, and Summers 2008; Gyourko, Hartley, and Krimmel 2021).

Table 2: TWFE estimates of regulatory responses

	Max. height (value) (1)	BCR (value) (2)	Denial probability (3)
FAR $\times$ Post	-0.82***	-0.02***	0.03***
SE	(0.18)	(0.01)	(0.01)
CI 95%	[-1.16 ; -0.47]	[-0.03 ; -0.01]	[0.02 ; 0.05]
R^2	0.45	0.44	0.05
N	4,857,093	4,298,907	87,115

Notes: Results from the estimation of Equation 1. Column 1 reports linear regressions where the dependent variable is the regulatory value at the parcel level: maximum heights (in meters, column 1) and BCR (ratio of building footprint to parcel size, between 0 and 1, column 2). Column 3 reports a linear probability model where the dependent variable is an indicator equal to 1 if the building permit was denied.. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2 Real estate dynamics

The reforms were intended to stimulate housing construction with the explicit aim of lowering prices and improving affordability. We therefore turn to real estate dynamics, examining the number of permits granted and built density (floor space per land area) at the zoning block level, and housing prices at the transaction level. Results for negative binomial regressions on permit counts are reported in Table 3. Results for density and prices are presented in Figures 4a and 4b (event studies) and in Appendix Table A.4 (main specifications).

Table 3 shows a small but statistically significant decline in the number of permits. The estimated coefficient of -0.049 implies an incidence rate ratio (IRR) of 0.95, meaning that the expected number of permits in treated areas is about 5% lower than in control areas. This effect is driven by a decline in approved permits, for which the estimates are negative and significant at the 1% level. This result is consistent with the earlier evidence of a significant increase in permit denials, with an IRR of 1.14.

We next examine whether the reduction in construction flows translates into changes in the housing stock by considering floor space density. Because construction takes time to translate into stock, we expect the effect on density to accumulate gradually rather

Table 3: Negative binomial estimations for permit counts

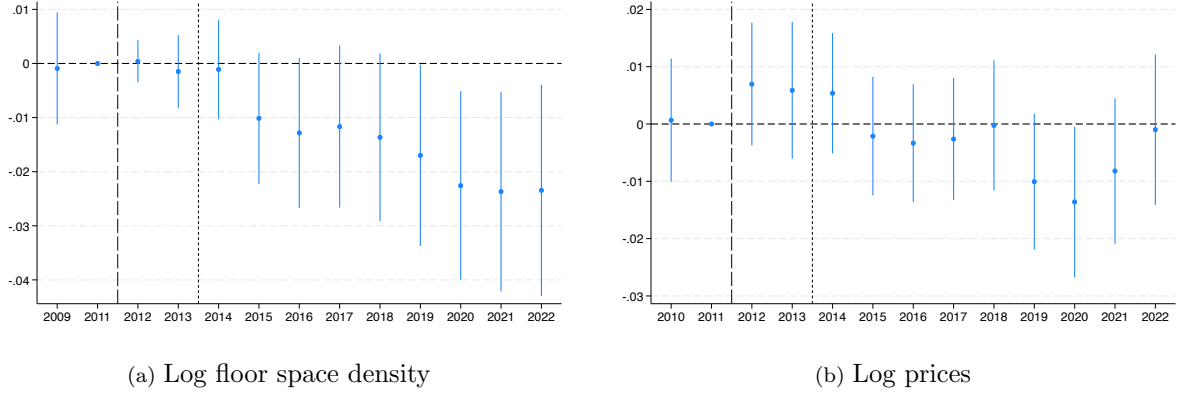
	Number of permits filed (1)	Number of permits accepted (2)	Number of permits denied (3)
FAR $\times$ Post	-0.049**	-0.098***	0.128***
SE	(0.021)	(0.026)	(0.037)
CI 95%	[-0.091 ; -0.007]	[-0.148 ; -0.048]	[0.055 ; 0.201]
N	122,445	122,445	122,445

Notes: Each column reports estimates from a negative binomial regression of Equation 3 at the zoning block level for the years 2007–2021. The reference year is 2011. Coefficients are reported with standard errors in parentheses, clustered at the zoning block level. All regressions include year and municipal fixed effects. Coefficients represent log changes in the expected count and should be exponentiated for interpretation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

than appear immediately. Consistent with this, the pooled TWFE estimate in Table A.4 averages a null early period with a growing later one, yielding a marginal -2% effect (10% level) that is not always robust to all specification (see column 3 in Table A.9). Nevertheless, the event-study estimates in Figure 4a and A.10a show why: the gap between treated and control areas is statistically indistinguishable from zero through 2018, then widens into a 2.4% gap by 2020, significant at the 95% level. The event-study estimates display an increasingly negative trend over time, consistent with a slow and modest adjustment of the housing stock in response to reduced construction flows. This does not imply a decline in density: treated neighborhoods continue to densify, but at a slower rate than control areas.

Construction may also have shifted spatially, with projects redirected from treated to control areas. In that case, the negative estimates for permits and density would partly reflect substitution rather than a net decline in construction. The pattern of results, however, argues against displacement as the primary explanation. If projects were simply redirected elsewhere, one would expect a decline in accepted permits but no systematic increase in denials: developers redirecting projects to other locations would not face rejection in the treated area. The simultaneous and significant increase in denied permits is instead the signature of active regulatory restriction. While we cannot fully rule out some displacement, and a direct test on permit trends in control areas would provide a cleaner bound, the denial evidence points to regulatory tightening as the main driver of the decline in construction.

Figure 4: Event-study estimates for real estate dynamics



Notes: Results from the estimation of event-study specifications. Panel a plots results using log floor space density as the dependent variable in Equation 4, modified to match the panel structure of the data, as in Equation 2. Panel b reports results for log price per square meter as the dependent variable. The event-study specification at the transaction level is $Y_j = \sum_{y \neq 2011} \beta_1^y \cdot FAR_{b(j)} \times 1_{t(j)=y} + \alpha_{m(j)} + \delta_{t(j)} + Z_j \cdot \gamma + \epsilon_j$, with Z_j a vector of hedonic controls including an indicator for single-family dwellings, floor space, number of rooms, and construction vintage. The reference year is 2011. Regressions include municipal and year fixed effects. Standard errors are clustered at the zoning-block level. Vertical bars represent 95% confidence intervals.

The evidence points to a modest decline in new construction and a small effect on overall built density. These outcomes are consistent with the regulatory reaction to upzoning documented above. Because there is no benchmark index of regulatory restrictiveness, we cannot determine whether overall restrictiveness returned to or exceeded its baseline level. In Section 7, we discuss mechanisms to provide insights to the scale of mayors' regulatory backlash relative to the initial deregulatory shock.

Finally, we examine the impact on housing prices. The dependent variable in Figure 4b is the log price per square meter, with additional controls for dwelling characteristics. Given the modest and partially offset decline in permits, standard supply–demand logic implies a negligible price effect. Using a housing demand elasticity of 0.4 for France (Chapelle and Trannoy 2023) and our largest supply variation estimand of 0.024, the maximum implied price effect would be 0.01, well below the detection threshold given our standard errors. Consistent with this prediction, point estimates are close to zero and not statistically significant. Price effects are more difficult to interpret than quantities. While land use regulation may affect supply locally, aggregate supply adjustments can influence prices in both treated and untreated areas. Our estimates therefore capture relative changes in neighborhood prices linked to land use regulation and built environment adjustments.

In short, the reforms triggered a regulatory backlash and failed to increase housing supply. If anything, the policy generated adverse consequences, leading to tighter land use regulation, a modest decline in new construction, and no measurable change in prices.

6 Robustness checks

Alternative level of fixed effects In appendix A.5.1, we report our main results using an alternative set of time invariant fixed effect by replacing municipality fixed effects $\alpha_{m(j)}$ with municipality–FAR-status fixed effects $\alpha_{m(j) \times FAR_b}$. Our results and conclusions remain virtually unchanged, confirming that they are not driven by unobserved heterogeneity specific to pre-reform FAR coverage within municipalities.

Exploiting within jurisdiction variations FAR may correlate with other time-varying policies implemented at different administrative levels, namely departments and municipalities. While departments play a limited role in urban planning and applied uniform stamp duty rates over the period, we nonetheless introduce department-by-time shocks to flexibly absorb any common real estate dynamics operating at that level. As shown in Appendix Tables A.8 and A.9, our estimates remain unchanged. More stringent concerns arise at the municipal level, where mayors may influence local development through taxation or complementary regulatory instruments. We therefore introduce municipality-by-time shocks to absorb any time-varying municipal-level factors. Under this specification, the estimated decline in maximum heights becomes slightly larger, suggesting that baseline estimates are, if anything, conservative. Results for BCR remain stable, though less precisely estimated due to reduced within-municipality variation. Permit denials and price effects are unaffected across specifications. For realized density, the baseline specification yields a negative and marginally significant estimate. Once municipality-by-time shocks are included, this effect becomes statistically insignificant. Overall, these results indicate that our findings are not driven by correlated policy changes at either the department or municipal level.

Distance-based sample restrictions To further address concerns about unobserved heterogeneity, we restrict the sample to observations located close to the border between treated and untreated zones. This approach compares areas exposed to similar local

shocks and sharing more similar unobserved characteristics, thereby strengthening the credibility of the parallel trends assumption. We progressively restrict the sample to treated and control units located within 500m, 250m, and 100m of the treatment boundary, corresponding to the 82nd, 63rd, and 33rd percentiles of the distance distribution. Appendix Table A.7 shows that results remain stable across all distance bands. For regulatory outcomes, point estimates increase in magnitude as the band narrows, with the 100m sample yielding the largest and most precisely estimated coefficients for both maximum heights and BCR. For permit denials, precision declines mechanically due to smaller sample sizes, but estimates remain consistent with the baseline results. Overall, these findings indicate that our results are not driven by differential local shocks or systematic differences between treated and control areas.

Controlling for observed differences We augment our baseline specification with baseline neighborhood-level characteristics interacted with time dummies. This allows for differential time paths across areas with distinct initial characteristics, relaxing the assumption that treated and control areas would have evolved similarly conditional only on fixed effects. Appendix Figures A.9 and A.10 reports the corresponding event-study estimates. The inclusion of these controls leaves both the magnitude and timing of the estimated effects virtually unchanged. Coefficient patterns remain stable across specifications, and pre-treatment trends remain statistically indistinguishable from zero. Our main results remain unchanged. These findings indicate that our results are not driven by differential observable characteristics between treated and control areas.

Alternative regulatory measurements Regulatory parameters are constructed from detailed zoning documents and may contain measurement noise, as multiple regulatory values can coexist within a given zone (e.g., height limits varying by street distance ; see Appendix A.2.2 for details). To ensure that our results are not driven by methodological choices, we construct alternative measures of regulatory restrictiveness based on threshold indicators. We build a dummy equal to one if regulation falls below a restrictiveness threshold, using 12 meters for maximum height and 0.3 for BCR following APUR documentation.¹² Appendix Figure A.11 reports the corresponding event-study estimates.

12. We focus on the intensive margin of regulation, namely changes in parameter values within zones where height and BCR rules already applied. Prior to the reform, these tools were nearly universal

Results remain qualitatively unchanged: treated areas become significantly more likely to be subject to restrictive regulatory values after the reform. The timing of the effect is sharp and concentrated after 2012, and confidence intervals are tighter than in the baseline specification. Overall, the dynamic pattern appears even cleaner than with continuous regulatory measures.

Timing of regulatory responses The removal of FAR followed an earlier nationwide upzoning reform that referred more broadly to “building rights.” Because FAR became central in the political debate, regulatory adjustments may have begun prior to the formal 2014 removal. In addition, the second reform was under preparation from late 2012 onward, raising the possibility of anticipatory responses.¹³ To assess this possibility, we group years into three periods: pre-reform (2011 and earlier), anticipation (2012–2013), and post-reform (2014 onward), and estimate a specification allowing for differential effects across these phases. Table A.11 shows that regulatory tightening began during the anticipation period and intensified after 2014. Height constraints tighten further in the post-reform period, while no additional adjustment is observed for building coverage ratios, which is consistent with building coverage can be constrained through multiple substitutable instruments (e.g., setback requirements). Permit denial probabilities increase only after 2014, consistent with effective denial becoming legally feasible once FAR were formally eliminated. Taken together, these patterns suggest that regulatory responses began prior to the reform and intensified thereafter, pointing to a gradual adjustment process rather than a discrete shift driven solely by the legal change.

7 Mechanisms

7.1 Motives of local elected officials

We next investigate why local elected officials responded to the reform by tightening other dimensions of land use regulation. This subsection focuses on two political mechanisms: partisan alignment with the central government and electoral pressure.

in treated areas, leaving limited scope for further adoption. Focusing on parameter stringency therefore provides a more credible basis for identification than variation in the mere presence of regulatory instruments.

13. Our identification remains valid if similar anticipatory adjustments occurred in non-FAR areas, as long as differential changes across treated and control zones reflect the removal of FAR.

Political alignment We first examine whether the regulatory backlash reflects partisan opposition to the central government. The reform was enacted by a left-wing central government. Municipalities governed by right-wing or centrist mayors can therefore be interpreted as opposition-led municipalities. If partisan alignment matters, these municipalities should have stronger incentives to offset the reform locally. Conversely, left-wing mayors may be less willing to counteract a reform promoted by their own political camp, either because they share its objective or because they face partisan incentives to support the policy. This mechanism is consistent with evidence that political alignment can improve policy implementation and government productivity (Spenkuch, Teso, and Xu 2023), and that alignment between local and higher-tier governments affects the allocation of intergovernmental resources (Solé-Ollé and Sorribas-Navarro 2008). However, existing evidence on partisan differences in local zoning policies remains mixed (Solé-Ollé and Viladecans-Marsal 2013; Ferreira and Gyourko 2023).

To test this hypothesis, we extend the baseline specification of Equation 1 by introducing interactions between treatment, the post-reform period, and a municipal characteristic $Z_{m(j)}$. The specification is:

$$\begin{aligned}
Y_j = & \alpha_{m(j)} + \delta_{t(j)} + \beta_1 \cdot FAR_{b(j)} \times 1_{t(j)>2011} \\
& + \beta_2 \cdot FAR_{b(j)} \\
& + \beta_3 \cdot FAR_{b(j)} \times Z_{m(j)} \\
& + \beta_4 \cdot Z_{m(j)} \times 1_{t(j)>2011} \\
& + \beta_5 \cdot FAR_{b(j)} \times 1_{t(j)>2011} \times Z_{m(j)} + \epsilon_j.
\end{aligned} \tag{5}$$

where $Z_{m(j)}$ denotes the municipal characteristic considered in each exercise. The coefficient β_1 captures the effect of the reform in municipalities with $Z_{m(j)} = 0$. The coefficient β_5 measures the additional effect in municipalities with $Z_{m(j)} = 1$, so that their total effect is given by $\beta_1 + \beta_5$. The coefficient β_2 captures the baseline relationship between pre-reform FAR constraints and the outcome in municipalities with $Z_{m(j)} = 0$, while β_3 allows this relationship to differ in municipalities with $Z_{m(j)} = 1$. The coefficient β_4 allows municipalities with $Z_{m(j)} = 1$ to follow a different post-2011 evolution independently of exposure to the reform. Because $Z_{m(j)}$ reflects pre-reform municipal characteristics, β_5 has a causal interpretation under the DiD identifying assumptions: it captures how pre-

existing local conditions amplified or attenuated the regulatory response to the reform, net of municipal and time fixed effects.

In the political-alignment exercise, $Z_{m(j)}$ is equal to one for municipalities governed by right-wing or centrist mayors in both 2008 and 2014, and zero otherwise. Table A.7.1 reports the results. The common coefficient remains statistically significant across outcomes, indicating that the regulatory backlash was not confined to opposition-led municipalities. The interaction coefficient provides little evidence that right-wing or centrist municipalities responded more strongly. If anything, permit denials increased less in these municipalities. These results suggest that partisan opposition to the central government was not the primary driver of the regulatory response.

Electoral pressure A second political mechanism is local electoral pressure. While central governments may seek to expand housing supply to address affordability concerns, local officials remain accountable to incumbent residents, who may oppose new development because of construction nuisances, congestion, neighborhood change, or concerns about property values. Mayors facing tighter electoral competition may therefore have stronger incentives to offset the reform. This interpretation is consistent with evidence that local housing policy responds to electoral incentives (Schmutz and Verdugo 2023).

To examine this mechanism, we estimate Equation 5 using an indicator equal to one when the mayor’s victory margin in the 2014 municipal election is below the sample median. The results, reported in Table 4, show that regulatory tightening is stronger in politically contested municipalities. In low-margin municipalities, maximum height falls more sharply, while permit denials increase more. BCR does not display a clear additional adjustment. These patterns suggest that mayors exposed to stronger electoral pressure were more likely to respond to the reform by restricting development.

Overall, political orientation alone does not explain the regulatory backlash. Tightening is observed across political orientations, and opposition-led municipalities do not respond more strongly in most specifications. By contrast, the response is stronger in politically contested municipalities. This points to local electoral incentives as a more plausible political driver than partisan opposition or party discipline.

These tests should be interpreted with some caution. The 2014 municipal elections

took place shortly after the law was enacted, so both political alignment and victory margins may partly reflect the political environment shaped by the reform itself. To address this concern, we also use alignment and victory margins measured in the 2008 municipal elections and find no evidence of heterogeneity. This is not fully conclusive, however, since the 2008 variables capture a different set of incumbents than those who implemented the regulatory adjustments, which mechanically attenuates any detected heterogeneity. A further limitation is that the reform was first proposed by a right-wing government and later enacted by a left-wing one, so partisan incentives may have shifted over the period. The triple-interaction coefficients β_5 are identified off pre-determined variation in $Z_{m(j)}$ and carry a well-defined causal interpretation under the DiD assumptions; what we cannot fully pin down is the exact channel through which a narrow margin operates. It most plausibly reflects mayors' heightened sensitivity to the preferences of incumbent residents, but we remain agnostic about the deeper mechanism.

Taken together, these findings deliver a simple message: sharing a policy platform with the central government, as proxied by partisan alignment, is not enough to offset local politicians' incentives to resist densification when their constituents oppose it. Local electoral accountability appears to dominate party discipline.

Table 4: Heterogeneity: victory margin below median in 2014

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-0.50*	-0.02***	-0.02***	0.03***	-0.01
SE	(0.18)	(0.27)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Margin (β_3)		-1.06*		0.03*		-0.05***
SE		(0.55)		(0.02)		(0.01)
FAR $\times$ Margin $\times$ Post (β_5)		-0.67*		-0.01		0.07***
SE		(0.35)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
250m bandwidth	N	N	N	N	N	N
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the land use regulation value at the parcel level as the dependent variable. The reference year is 2011. Values for maximum heights are in meters. Values for BCR are expressed as ratios between building footprint and parcel size, ranging from 0 to 1). In this table, $Z_{m(j)}$ is equal to one for municipalities where the mayor’s victory margin in the 2014 municipal election is below the sample median. The common coefficient corresponds to β_1 . The baseline relationship between pre-reform FAR constraints and the outcome corresponds to β_2 . The baseline difference in this relationship for municipalities with $Z_{m(j)} = 1$ corresponds to β_3 . The differential post-2011 evolution for municipalities with $Z_{m(j)} = 1$ corresponds to β_4 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect in municipalities where the mayor was elected with a narrow margin in 2014. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block-level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

7.2 Motives of voters

The previous section shows that electoral incentives shape mayors’ responses to the reform. This raises the question of which local interests drive this resistance. A central mechanism in the literature is the “homevoter” hypothesis: homeowners oppose densification because nearby development may reduce neighborhood amenities and threaten property values, and they have the local voting power to influence land-use decisions (Fischel 2001). Yet homeownership may proxy for two distinct forces: the political weight of owner-occupiers, and the financial value of the assets potentially affected by new development.

Distinguishing these forces is empirically difficult. In the U.S. context, low-density

neighborhoods with high amenities often coincide with high homeownership rates and high housing values. Existing evidence is therefore mixed: some studies find that municipalities with higher homeownership rates adopt stricter land-use regulations (Hilber and Robert-Nicoud 2013; Mast 2024; Dobbels and Tavakalov 2024), while others find no such correlation (Jedwab, Barr, and Brueckner 2022).¹⁴ In France, this distinction may be particularly relevant because property ownership and local residence do not perfectly overlap, and landlords often invest in their own municipality of residence (Levy 2022). Moreover, where property values are high, owners have more to lose from nearby development and may therefore face stronger incentives to oppose or litigate against new projects (Spier 2007; Lafont and Saisset 2026; Jones 2026).

To investigate whether homeowner electoral pressure shapes regulatory responses, we compare municipalities across varying homeownership rates. Under the “homevoter” hypothesis, resistance to densification should be more pronounced where homeowners constitute a larger share of the electorate. We test this mechanism by interacting the treatment with a dummy variable indicating whether the municipality’s homeownership rate is above the sample median. The results, presented in Table A.7.3, reveal no significant association between this interaction term and our three regulatory outcome measures. Similar null results are obtained when we use alternative proxies for attachment to low-density development, namely indicators for whether the share of dwellings that are detached houses or the share of municipal land that remains open or undeveloped is above the sample median (Tables A.7.5 and A.7.4). Our empirical setting therefore provides limited support for the hypothesis that regulatory responses are intensified by higher concentrations of homeownership or detached housing.

We then turn to housing prices, which provide a more direct measure of the property values at stake. When we interact the treatment with an indicator for municipalities whose housing prices are above the sample median, we find a stronger response in high-price municipalities (Table A.7.6). This pattern is consistent with a property-value exposure mechanism: resistance to densification appears stronger where the value potentially threatened by new development is higher. This interpretation is consistent with the idea that owners have stronger incentives to engage in costly opposition when the assets they seek to protect are more valuable (Spier 2007), and with recent evidence by Lafont and

14. See Hilber, Koster, and Loumeau (2024) for a comprehensive review.

Saisset (2026) showing that litigation against building permits is more frequent in high-value municipalities. It is also in line with Gyourko and McCulloch (2024), who find that aversion to density is concentrated in affluent, low-density neighborhoods. Finally, we examine heterogeneity with respect to municipal demographic structure, namely income, age, and the share of social housing, using indicators equal to one when the corresponding municipal variable is above the sample median. We find no systematic or coherent association with the regulatory response (Tables A.7.9, A.7.7, and A.7.8 in Appendix).

Taken together, the two positive findings of stronger responses in politically contested municipalities and in high-price municipalities are mutually consistent: where property values are high, residents have greater financial stakes in preventing densification, which plausibly strengthens electoral pressure on mayors to offset the reform. The null results on homeownership rate and demographic structure suggest that financial exposure, rather than the mere presence of owner-occupiers, is more closely associated with local regulatory resistance. In our setting, homeownership and property wealth are therefore not interchangeable. Homeownership may proxy for the financial stakes associated with densification, but property values capture these stakes more directly; consistent with this interpretation, regulatory resistance is concentrated in high-price municipalities rather than in municipalities with higher homeownership rates.

8 Conclusion

This paper studies how local governments respond to a centrally imposed reform when policy authority is shared across levels of government but political incentives are misaligned. Exploiting the nationwide abolition of maximum floor area ratios in France, we show that municipalities strategically adjusted other regulatory instruments and permitting practices in ways that offset the reform’s intended effects. Despite the removal of a binding density constraint, relative density growth slowed significantly in treated areas by the end of our sample, while housing prices showed no detectable change. Policy outcomes therefore reflect strategic local responses rather than the mechanical effect of deregulation.

We investigate the sources of this regulatory backlash. We find little evidence that partisan alignment between central and local governments plays a meaningful role. In-

stead, regulatory tightening is significantly stronger in municipalities where incumbent mayors were elected with a thinner margin and when property prices are higher. These patterns point to local electoral incentives as the operative mechanism: incumbent mayors reallocate restrictiveness across the remaining regulatory instruments to preserve the effective density constraint, in response to pressure from high-value property owners rather than from a homeownership-based backlash.

Our results carry a direct lesson for the design of centralized upzoning mandates. Relaxing a single density constraint accomplishes little when local governments retain the instruments and the incentives to restore it. Reforms that aim to expand housing supply from the center must therefore either constrain the full set of local regulatory margins or realign local incentives, for instance by compensating the communities that accommodate new development.

More broadly, our findings have implications on decentralization. When the authority remains partially shared, centralized reforms justified by cross-jurisdictional externalities may trigger strategic responses to protect local preferences. When the authority is partially shared, the existence of cross-jurisdictional externalities create a pressure toward the upward transfer of regulatory prerogatives to higher tiers of government. This consideration is likely to extend beyond land use regulation to other domains in which implementation authority is fragmented across levels of government, such as education or environmental policy.

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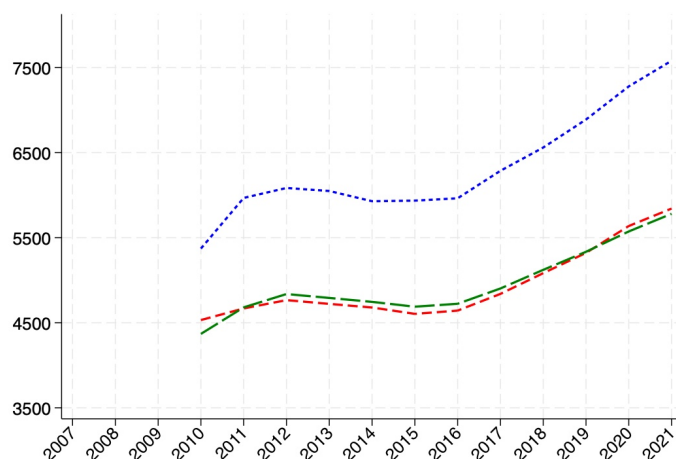
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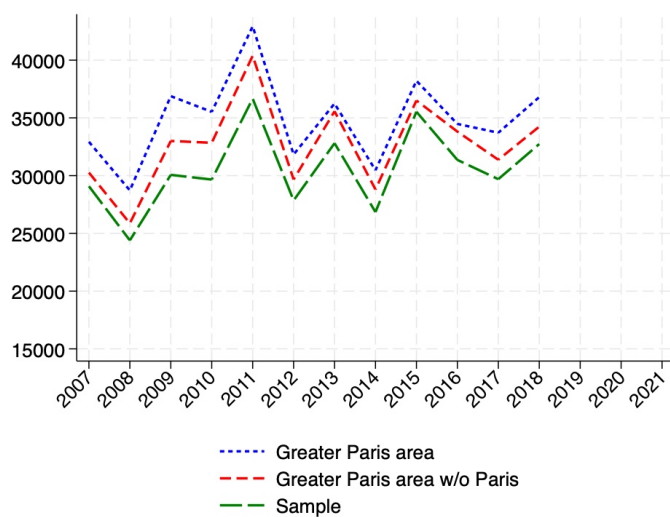
Appendix

A.1 Additional figures and anecdotal evidence

A.1.1 Housing markets



(a) Price per square-meter by year (€)



(b) Number of new dwellings authorized by year

Figure A.1: Real estate dynamics in the Greater Paris area (2007-2021)

Notes: Authors' calculations from DV3F (Panel a) and Sit@del2 (Panel b).

Table A.1: Comparison between Paris urban area and sample

	PUA	PUA (w/o Paris)	Sample
Surface (sq. km.)	679	613	567
Parcels	709,371	631,671	591,349
Developed Parcels	540,541	471,807	440,986
Dwellings	3,631,989	2,213,761	2,095,889
Houses	420,629	414,262	385,047
Apartments	3,211,360	1,799,499	1,710,842
Apartments (%)	88.4	81.3	81.6
Social Housing	717,517	558,426	521,690
Social Housing (in %)	19.8	25.2	24.9
Homeowners	1,355,380	876,404	831,624
Homeowners (in %)	37.3	39.6	39.7
Inhabitants	7,179,807	4,889,583	4,613,751
Household Size	2.0	2.2	2.2
Age of household head	52.2	52.2	52.2
Household income	46,844	41,088	41,572
Income per capita	23,291	19,239	19,490
Building Permits (2009–2021)	73,845	63,482	59,892
New Dwellings Authorized (2009–2021)	501,830	450,524	425,355
Transactions (2010–2021)	1,044,825	679,863	647,614
Average Price (2010–2021)	392,132	328,276	333,169
Average Price per sqm (2010–2021)	6,576	5,080	5,159

Notes: Authors' calculations. Paris urban area as defined by INSEE. Average value of the period 2007–2021, unless stated otherwise. Data sources: Fichiers fonciers (lines 2 to 7) / Filocom (lines 8 to 18) / Sitadel (lines 19 and 20) / DV3F (lines 21 to 24).

A.1.2 Building envelopes

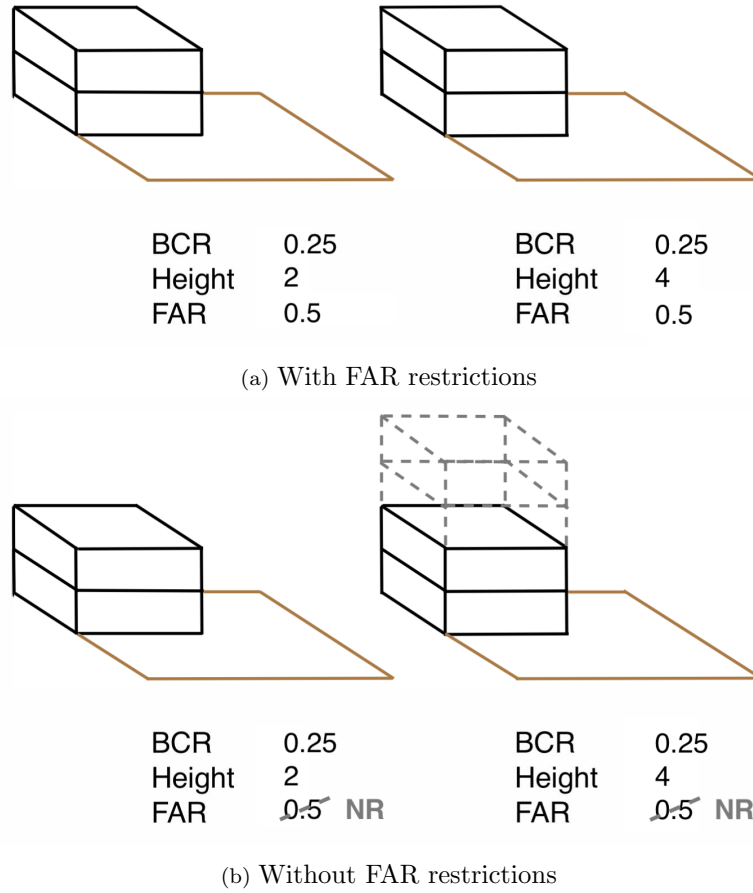


Figure A.2: Examples of regulatory building envelopes

Notes: This figure illustrates the effect of removing maximum FAR requirements on the building envelope (“building rights”). The building envelope is the three-dimensional space defined by zoning regulations within which construction is permitted on a given lot. Maximum height restrictions limit the possible number of floors. By construction, $FAR = BCR \times N_f$, where N_f denotes the number of floors. The figure provides a visual example of the interaction between the tools. If $FAR > BCR \times N_f$, FAR was not binding, and removing it had no effect on building envelope (right panel). In contrast, if $FAR \leq BCR \times N_f$, FAR was binding, and its removal increases building envelope (left panel). In the top panel, both parcels are subject to the same BCR and FAR but differ in maximum height requirements. The resulting building envelopes are identical. The removal of FAR requirements, as shown in the bottom panel (“NR”, for not regulated), expands the building envelope only for the right-hand side parcel because FAR was binding. For the left-hand side parcel, FAR was redundant with the BCR and maximum height requirements.

A.1.3 Cadastral data

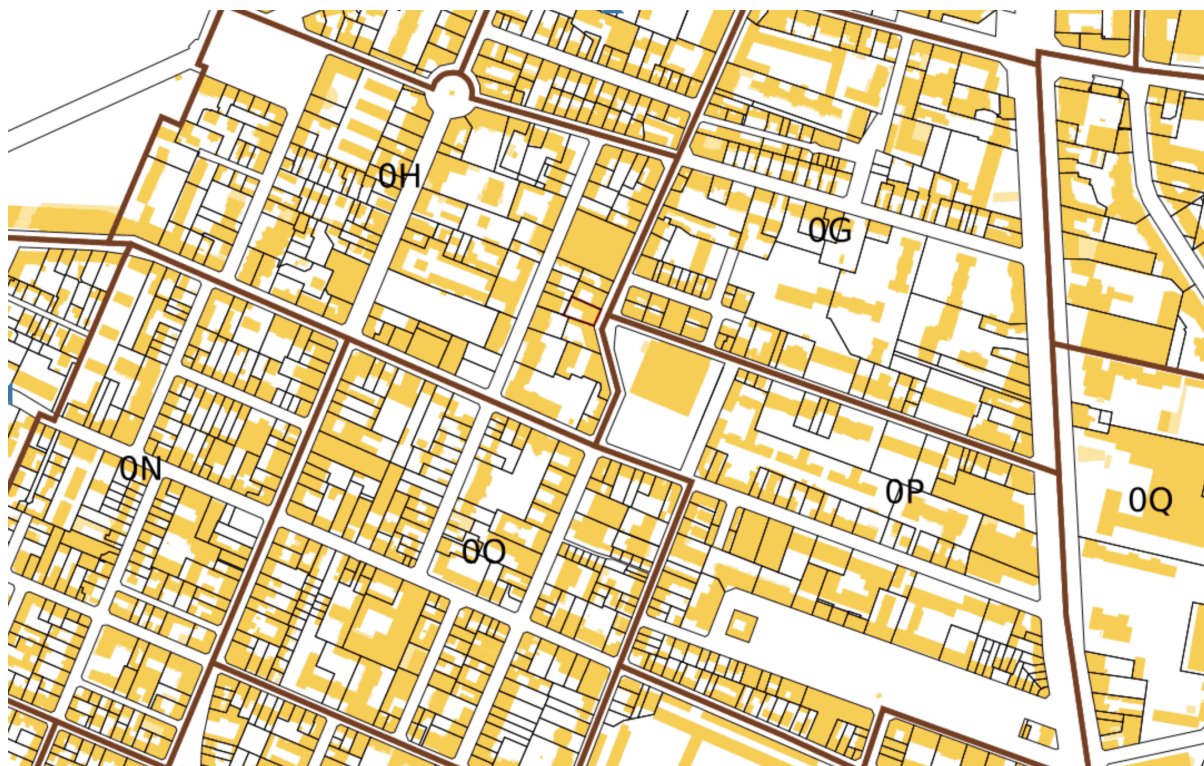


Figure A.3: Illustration of blocks and parcels in the French *cadastre* (land registry)

Notes: This figure shows examples of fiscal blocks and parcels in the French *cadastre*. The illustration considers blocks within the municipality of Montrouge, in the Paris area. Blocks labelled 0G, 0H, 0N, 0O, 0P and 0Q are fiscal blocks (*sections cadastrales* in French). Subdivisions within each block are parcels (*parcelles* in French). Source: screenshot from www.cadastre.gouv.fr.

A.1.4 The 2012 presidential race and the ALUR Act

Reducing land use restrictions to stimulate construction emerged as a bipartisan issue during the 2012 French presidential election. Both Nicolas Sarkozy, the conservative incumbent, and François Hollande, his socialist challenger, acknowledged the necessity of easing building regulations to address housing shortages. However, their respective legislative approaches diverged significantly regarding the balance of power between local and national authorities.

In March 2012, President Nicolas Sarkozy enacted a law mandating a uniform 30% increase in building rights, giving mayors six months either to adapt their local regulations or to opt out by council vote. Crucially, the law did not specify how this increase would translate into concrete changes in zoning rules, leaving implementation ambiguous. The law faced immediate resistance from mayors, who argued that a one-size-fits-all standard ignored local market conditions. It was also criticized as legally fragile, procedurally burdensome, and redundant with existing but rarely used bonus provisions.¹⁵

By contrast, Hollande’s government introduced the *Accès au Logement et Urbanisme Rénové* (ALUR) Act in March 2014, which curtailed municipal autonomy by abolishing FARs and minimum lot sizes and by mandating transfers of planning authority to intermunicipal entities (*Etablissements Publics de Coopération Intercommunale*, EPCI), though municipalities could block such transfers by council vote. In practice, no municipality in our sample adopted an intermunicipal plan during the study period, except Est Ensemble (Bagnolet, Bobigny, Bondy, Le Pré Saint-Gervais, Les Lilas, Montreuil, Noisy-le-Sec, Pantin, Romainville) in 2020. More details on the effects of intermunicipal cooperation can be found in Tricaud (2025). The key distinction between Sarkozy’s and Hollande’s initiatives lay in their treatment of local authority: The first law preserved municipal discretion, while the ALUR Act directly constrained it, in line with Hollande’s

15. See Valérie Valin-Stein, “Focus sur la majoration de 30 % des droits à construire: construire plus... mais sous conditions” *Le Particulier*, 04/01/2017. The article reports that: “Some homeowners have already begun to assess the potential benefits of this law. ‘We’ve had clients come in who wanted to increase the sale price of their home by 30% on the grounds that the right to build could be increased by the same amount. [...] In so doing, they benefit from the leverage effect of building for €1,500 per square meter (the average cost of construction) in municipalities where a square meter of living space can be negotiated at up to €8,000. [...]’ explains Laurent Demeure, President of Coldwell Banker France and Monaco. ‘When applied, this increase may facilitate the division of plots,’ points out Stéphanie Le Beuze, founder of Terra in design [...]” The article also highlights the cold reception of the law by local officials: “Mayors seem rather reluctant to increase the density of their municipalities. [...] The biggest obstacle is undoubtedly the outcry that elected representatives are likely to face. ‘Increasing density means running the risk of neighborhood problems,’ points out Gilles Bayon de la Tour.”

broader agenda to reduce France's political fragmentation across its 35,000 municipalities. Nevertheless, municipalities continued to wield substantial influence over land use regulation, as they could opt out of intermunicipal cooperation and retained broad discretion over many other instruments.

A.2 Data construction

A.2.1 Source data

Our estimation samples combine one original dataset, the geospatial panel of land use regulation, with several administrative sources. We summarize the provenance of each below; the construction of the regulation panel itself is detailed in Section A.2.2.

Land use regulation (*Bâtir Technologies* and APUR) We recover zoning for two benchmark years. For 2014, we rely on a unique source compiled by *Bâtir Technologies*, a firm that helps developers identify feasible construction projects, whose scanned PDF maps were georeferenced and vectorized by our research assistants. For 2018, the *Atelier Parisien d’Urbanisme* (APUR), the Parisian urban planning agency, directly provided vectorized zoning maps. The regulatory provisions are taken from standardized *Bâtir Technologies* documents, which record modifications to provisions along with their dates.

Land registry (*Fichiers fonciers*) The *Fichiers fonciers*, the French land registry (*cadastre*), divide land into municipalities, cadastral sections (housing blocks), and parcels, and are available annually from 2009 onward except 2010. Beyond parcel geolocation, they provide detailed information on owner characteristics, land use, occupancy status, and building features derived from fiscal data, which we use to track the housing stock.

Building permits (Sit@del2) Produced by the Ministry of Housing, Sit@del2 centralizes all building permits issued in mainland France. It reports newly authorized and initiated housing projects, including parcel location, application date, and project characteristics such as number of units, total built area, and lot size, as well as the administrative decision (authorized or denied).

Real estate transactions (DV3F) DV3F, provided by CEREMA, is compiled by the fiscal administration from two main sources: real estate transaction records collected by the land registry and notaries, and the *Fichiers fonciers*. It covers all transactions in France and reports details such as the transaction date, deed type (sale, exchange, auction), and sale price, providing a comprehensive record of property transfers.

Residential occupancy (FILOCOM) FILOCOM (*Fichier des Logements par Commune*) is a large-scale administrative database on residential occupancy across municipalities, compiled from tax records and other government sources. It provides granular information on dwellings and occupants, including household income and composition, occupancy status, and geolocation at the parcel level (Chapelle, Gobillon, and Vignolles 2022).

A.2.2 Panel of land use regulation

A.2.2.1 Sources

Each plan divides the territory into zones that are subject to the same regulatory provisions. To construct the regulation panel, we build zoning maps that we associate with the corresponding regulatory provisions.

- A. **Zoning maps.** We recover zoning for two benchmark years, 2014 and 2018. For 2014, we use scanned zoning maps from *Bâtir Technologies*. These maps were georeferenced and vectorized by our research assistants. For 2018, APUR directly provided vectorized zoning maps.
- B. **Regulatory provisions.** We collected regulatory documents from *Bâtir Technologies*. These standardized documents record modifications to provisions along with their respective dates. For each zone, there are 14 numbered articles, preceded by a preamble describing general zoning orientations. Each article corresponds to a specific regulatory dimension, as follows:

1. Prohibited Land Uses And Occupations
2. Land Uses And Occupations Subject To Special Conditions
3. Access And Roads
4. Servicing By Networks
5. Lot size and characteristics
6. Building Setbacks In Relation To Roads And Public Easements
7. Building Setbacks In Relation To Property Boundaries
8. Building Placement Relative To One Another On The Same Property

9. Building Coverage Ratio
10. Building Height
11. Exterior Appearance – Site Protection
12. Parking requirements
13. Open Spaces And Landscaping
14. Floor Area Ratio

A.2.2.2 Construction of the panel

The final panel of land use regulation combines two intermediate datasets: One describing zoning provisions and the other linking parcels to zoning codes.

Step 1: Panelization of regulatory rules We encode the content of the zoning plans. For each zone within a municipality, four provisions can be systematically measured: minimum lot size, maximum building coverage ratio, maximum building height, and maximum floor area ratio. In doing so, we faced two main challenges.

The first challenge is that multiple values often coexist within a single zone, especially for maximum heights. We record both the minimum and maximum values and use their average as the reference, excluding annexes (e.g., garages) and decorative elements. To ensure comparability, we standardize on ridge height (or total height for flat roofs), applying conversions where needed. Robustness checks using minimum and maximum values separately yield similar results (available upon request).

The second challenge is that some provisions are difficult to interpret because they depend on external “master plans” or relative measures tied to lot characteristics. For example, maximum heights can be expressed as absolute meters above sea level rather than relative to ground elevation. Master plan provisions typically authorize large, negotiated projects ex post. Such zones cover only 12 km² of our final sample (versus 324 km² regulated by FAR). Our treatment specification excludes these zones, though including them yields similar results (not shown).

This process yields a dataset containing:

- Zoning code (municipality specific zone identifier)
- Year of modification

- Regulatory provisions (FAR, BCR, maximum height, minimum lot size)

We then construct a panel at the zone–municipality–year level. Missing years (i.e., years without plan modifications) are filled using the most recent available regulation. For example, if a plan is observed in 2010 and again in 2014, the 2010 provisions are carried forward to 2011–2013. This yields an unbalanced panel, as zones remain in the dataset only while they exist; rezoning can remove existing zones and create new ones.

Step 2: Construction of the zoning plans panel The 2018 zoning plan, vectorized by APUR, is directly usable. To construct the 2014 counterpart, we first geocode the PDF maps by identifying reference points within each municipality. A georeferencing algorithm then converts the PDFs into raster files. We manually digitize polygons to delimit zone boundaries and assign their names.

We combine zoning provisions and maps with the French cadastre as of January 1, 2014 to obtain a stable unit of observation over time. Using a single reference year avoids the attrition that would arise if parcels were re-matched each year, since parcel reallocations (splits or mergers) change identifiers, making it impossible to track parcels longitudinally. In addition, reallocations are frequent and may be non-random. This matching procedure yields an initial cross-sectional dataset of 555,059 parcels. We panelize the data for 2007–2021 as follows:

- Each 2014 parcel is matched geographically with both the 2014 and 2018 zoning maps using PostGIS, yielding two observations per parcel.
- For missing years between 2007 and 2021, we assign zoning codes according to these rules:
 1. For years before 2014, parcels retain their 2014 zoning code.
 2. For years after 2018, parcels adopt their 2018 zoning code.
 3. For years between 2014 and 2018:
 - If the zone code did not change (382,459 parcels), we assign the same code.
 - If the code differs (172,600 parcels), we use the regulatory provision panel to identify the year of change. If a 2018 zone code did not exist in 2014, we assume it was introduced through rezoning changes between 2014 and

2018. We then use the regulatory provision data to identify the first year this code appears and assign it to the parcel from that year onward. For example, if a parcel receives a new code absent in 2014 but first recorded in 2016, we assign the 2014 code up to 2015 and the new code from 2016 to 2018.

Step 3: Creating a panel of land use regulation Finally, we match the parcel panel with the regulatory provision panel by zoning code and year. The resulting unbalanced panel contains 482,912 parcels across 118 municipalities. Figure A.4 maps their spatial distribution, while Table A.2 reports attrition rates.¹⁶

Parcels in the full panel appear in light green, while those in the cylindrical (balanced) panel are shown in dark green. White areas represent territories outside Greater Paris or municipalities without recoverable zoning data. Grey areas correspond to parcels that could not be matched between zoning maps and regulatory provisions. Two situations explain most of these cases: (i) non-urban zones (natural, to-be-urbanized, or agricultural) and (ii) (sub-)zoning mismatches. In practice, zones are often subdivided into subzones (e.g. UA into UAa and UAb). Parcels cannot be matched when one dataset reports only the general zone while the other provides detailed subzones. It applies to areas regulated through master plans, for which zoning plans lack detailed sub-zoning. In our main sample, we further exclude all parcels zoned as natural or agricultural areas, and areas regulated through master plans (see Figure 1).

Paris is excluded from the analysis. Its housing market is very specific and lacks a close empirical counterpart. Its political structure is also unique: Paris is both a municipality and a department, with mayors appointed by arrondissement councils rather than directly elected. Additionally, Paris used a uniform FAR across the city (except for railways and certain public or historic parcels), while our identification relies on within-municipality variation.

16. The quality of the match between rules and zones declines with distance from the observation year.

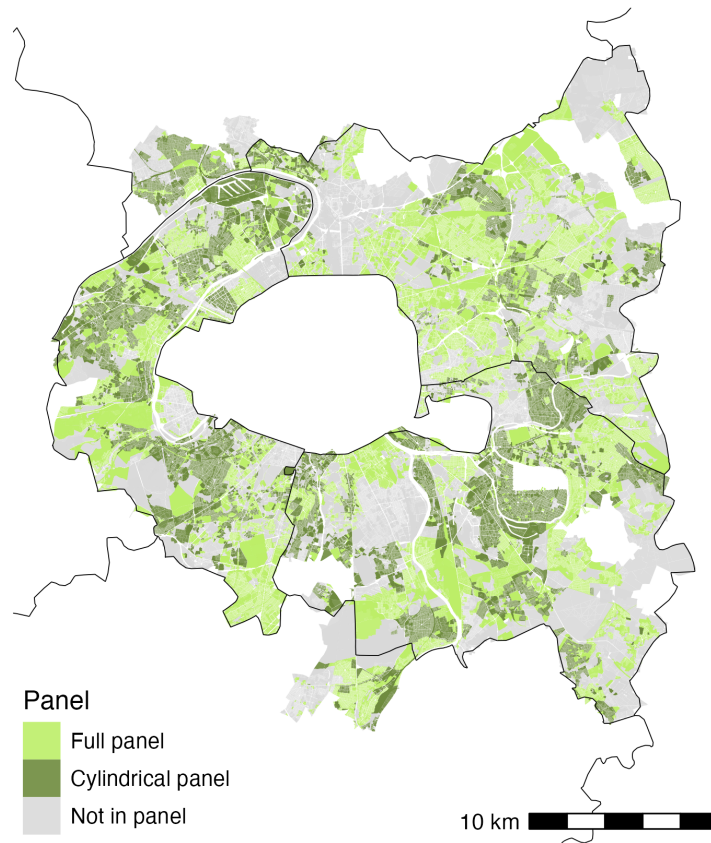


Figure A.4: Spatial distribution of observations in the panel (2014)

Notes: The figure shows all parcels in our sample as of January 1, 2014, including non-urban zones. Dark green indicates parcels observed in every year of the period 2007–2021. Light green indicates parcels observed at least in 2014. Grey indicates parcels that could not be matched between zoning maps and regulatory provisions. White indicates areas outside the scope of the study, either by sample design or due to missing zoning data.

Parcels by year		Parcels by frequency	
Year	N	Frequency	N
2007	299,367	0	4,408
2008	325,537	1	4,236
2009	336,427	2	7,213
2010	367,842	3	10,273
2011	387,580	4	11,196
2012	397,991	5	18,428
2013	435,935	6	18,241
2014	427,352	7	29,274
2015	413,753	8	49,712
2016	394,311	9	33,001
2017	379,661	10	20,947
2018	365,743	11	28,858
2019	364,959	12	26,901
2020	342,546	13	30,801
2021	346,958	14	189,423
		Total	482,912

Table A.2: Panel structure distribution

Notes: Table of parcels by year (left) shows the number of observations in the parcel data for each year of our observation period (2007-2021). The years for which we recover the most parcels are 2013 and 2014, consistent with the fact that we use the zoning map division of 2014. Table of parcels by frequency (right) results from the computation of frequency of parcels identifier in the full (unbalanced) panel data of parcels over the period. 37% of observations appear every year (cylindrical panel), for a total of 501,081 distinct parcel identifiers.

A.3 Descriptive statistics

A.3.1 Regulations

A.3.1.1 In 2014

Our final panel includes 118 municipalities. Among them, only 17 did not use FAR in any part of their master plan as of early 2014. The remaining 101 regulated new constructions using FAR, but none applied it in all zones. The weighted distribution of regulatory provisions is provided for 2014 in Table A.3. Figures A.5 show the spatial distribution of regulatory requirements.

FAR covered 52% of the study area, and 50% excluding master plans. In 2014, FAR values are set between 0.02 and 3.5, with an average of 0.78 and a median of 0.48. When only one value is set, the restrictiveness is in general greater, with a mean FAR at 0.72 and a median at 0.6, which is in line with the idea that it tolerates fewer exceptions and is hence more restrictive. This conclusion also applies for BCR and maximum height. Building coverage ratios are between 0.05 and 1. Mean and median BCR are between 0.4 and 0.5. The average maximum height is 14.7 m, with a median of 13. The average minimum lot size is 638 m² with a median of 450 m². For reference, across all 2014 parcels in the study area, the average lot size is 1359 m² and the median is 340 m² and 50% of parcels are between 195 and 543 m².

Table A.3: Land use regulation descriptive statistics, as of January 1, 2014

	Mean	SD	Min	p25	Median	p75	Max	N	Surface (km^2)
FAR									
Dummy	0.52	0.50	0	0	1	1	1	435,724	324
Dummy (w/o master plan)	0.50	0.50	0	0	1	1	1	421,118	312
Average	0.78	0.48	0.02	0.43	0.65	1	3.50	260,097	175
Single value	0.72	0.49	0.02	0.40	0.60	1	3.50	157,695	112
Min value	0.60	0.33	0.05	0.40	0.50	0.70	2.50	105,206	65
Max value	1.18	0.64	0.25	0.70	1	1.50	3.20	105,206	65
BCR									
Dummy	0.79	0.40	0	1	1	1	1	435,724	324
Average	0.51	0.16	0.05	0.40	0.50	0.60	1	347,569	254
Single value	0.46	0.17	0.05	0.40	0.40	0.60	1	175,108	132
Min value	0.44	0.16	0.10	0.40	0.40	0.50	0.90	178,255	120
Max value	0.68	0.17	0.20	0.60	0.65	0.80	1	178,255	120
Height limit									
Dummy	0.96	0.19	0	1	1	1	1	435,724	324
Average (m)	14.67	5.95	3	10	12.50	18	51.50	392,875	292
Single value (m)	14.10	6.12	3	10	12	16	50	231,278	174
Min value (m)	12.71	5.32	3	10	11	15	50	164,383	120
Max value (m)	18.30	6.66	8	12	16.50	22.50	62	164,383	120
Lot size									
Dummy	0.24	0.43	0	0	0	0	1	435,724	324
Average	638	467	150	300	450	800	3000	134,143	77
Single value (m^2)	563	479	150	300	375	600	3000	84,981	44
Min value (m^2)	660	1600	150	300	300	400	10000	52,886	33
Max value (m^2)	1119	732	250	500	800	1600	4000	52,886	33

Notes: We use 2013 data, reflecting regulations as of December 31, 2013 / January 1, 2014. Column 1 reports the number of observations per variable, and Column 2 the total surface covered by each of the four regulatory tools: FAR, BCR, maximum height, and minimum lot size. Columns 3–8 provide distributional statistics of values, weighted by zone surface. For each tool, we first compute a dummy equal to 1 if the tool is used. All other rows report weighted means. When zones have multiple values for the same tool, our baseline measure is the average between the minimum and maximum (shown as “Average”). We also report weighted means based on zones with a single value only (“One value”), on the minimum (“Min value”), and on the maximum (“Max value”).

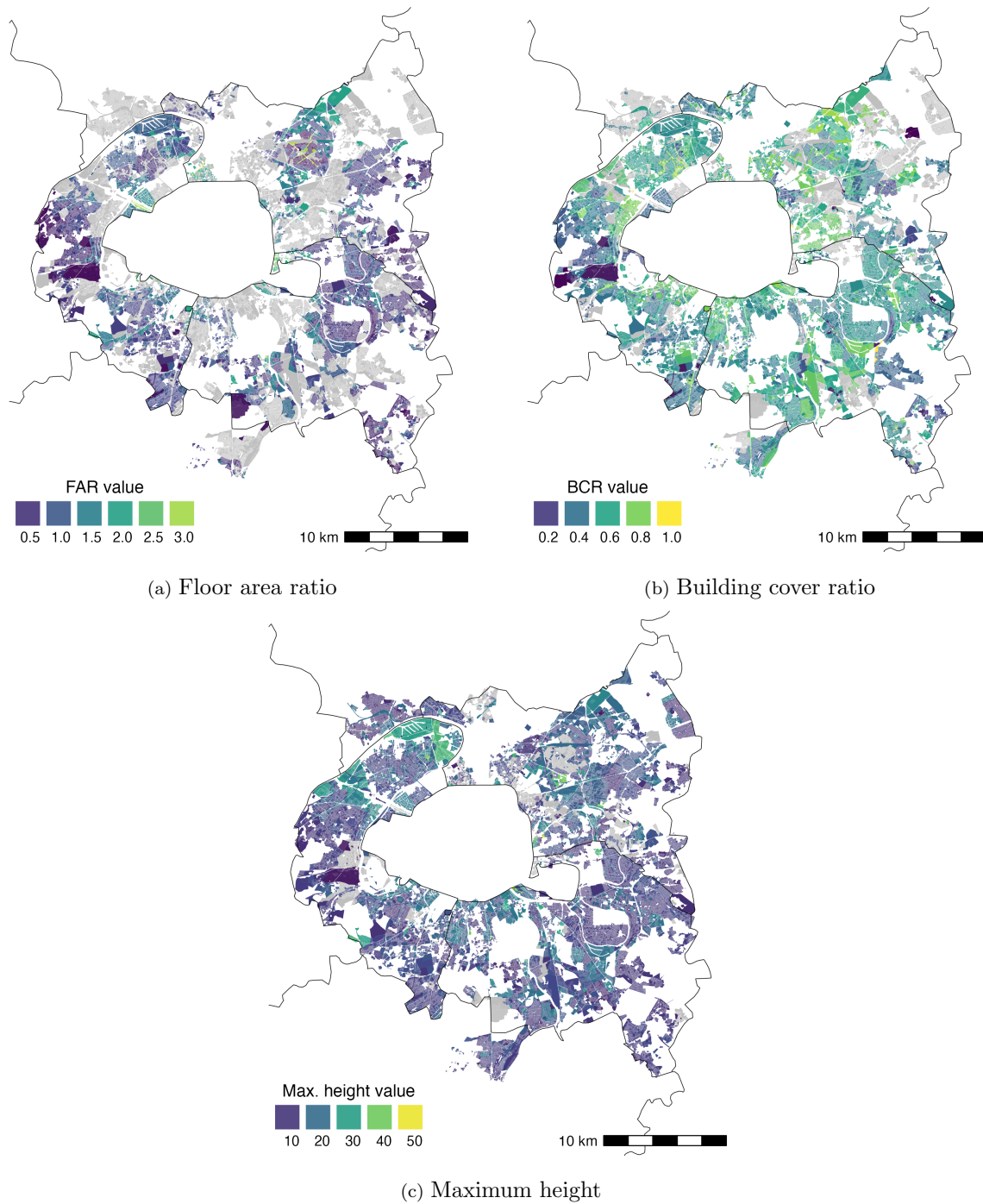
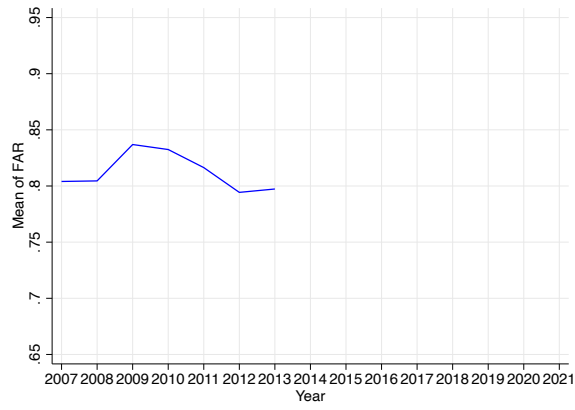
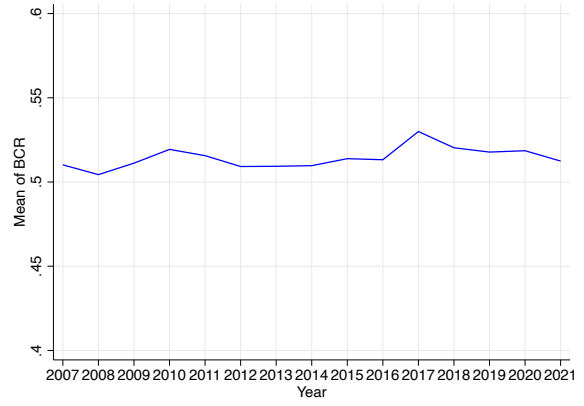


Figure A.5: Areas covered by different land use measures in 2014

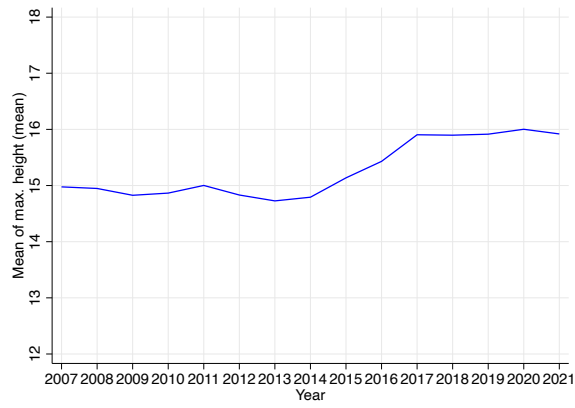
Notes: Figures A.5a, A.5b, and A.5c plot the zones covered by, respectively, maximum FAR, maximum BCR, or maximum height requirements as of early 2014 in the Paris urban area. Light grey areas are zones in our sample that did not use the corresponding tool to regulate land use. White areas are outside the scope of our study, either because of non-residential uses (green spaces, airports), sample choice, or missing data.



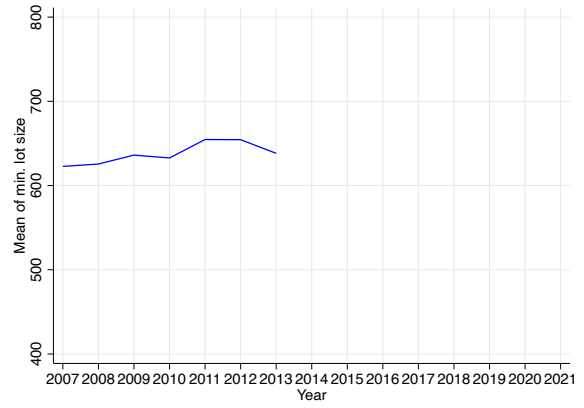
(a) Floor area ratio



(b) Building coverage ratio



(c) Maximum heights



(d) Minimum lot sizes

Figure A.6: Evolution of regulatory values (2007-2021)

Notes: Figures A.6a, A.6b, A.6c, and A.6d show the evolution in the mean of, respectively, FAR, BCR, maximum height, and minimum lot size requirements, weighted by regulatory zone size. Values are stable across the period, except for maximum heights.

A.3.1.2 Minimum lot sizes

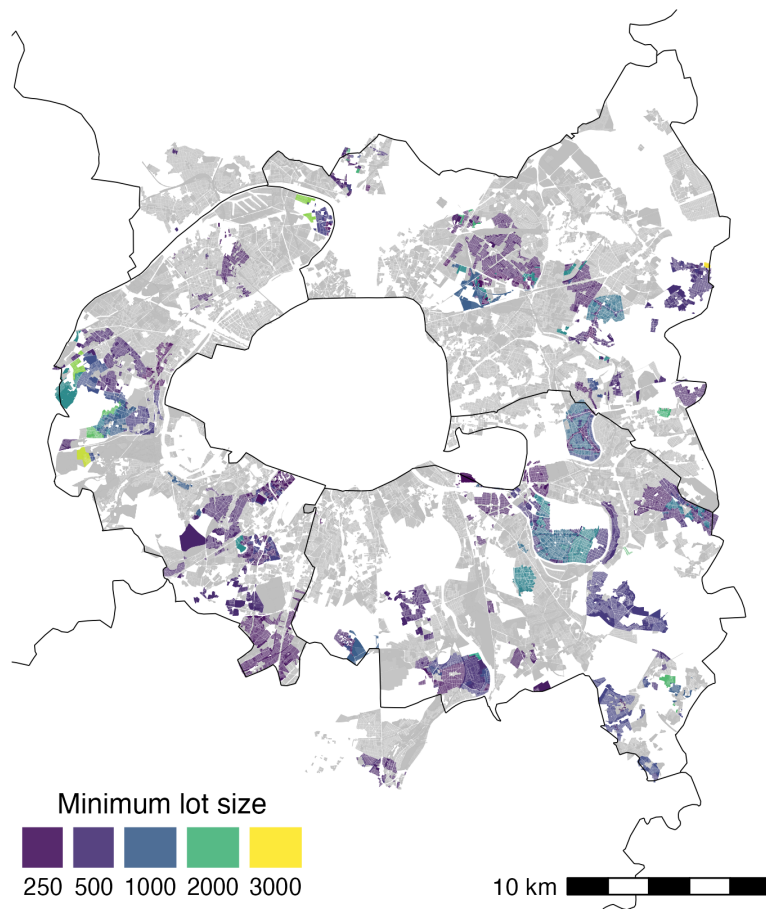
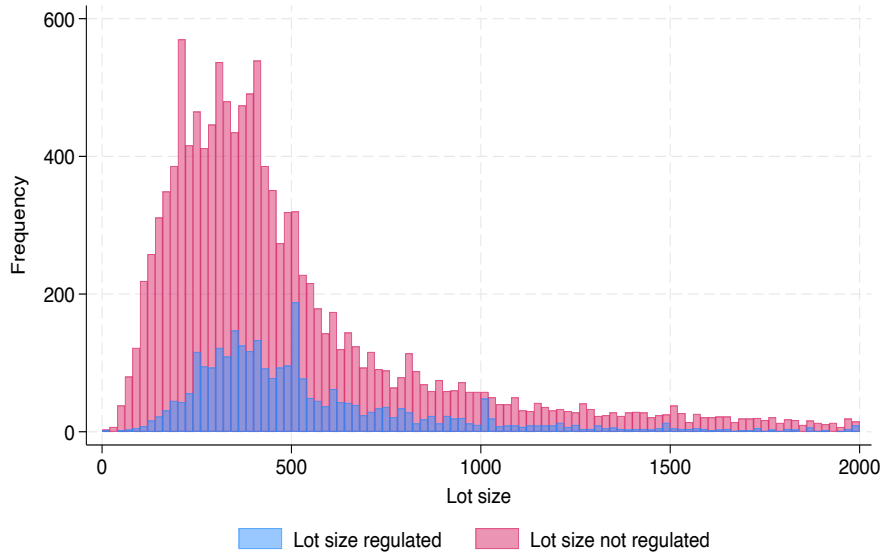
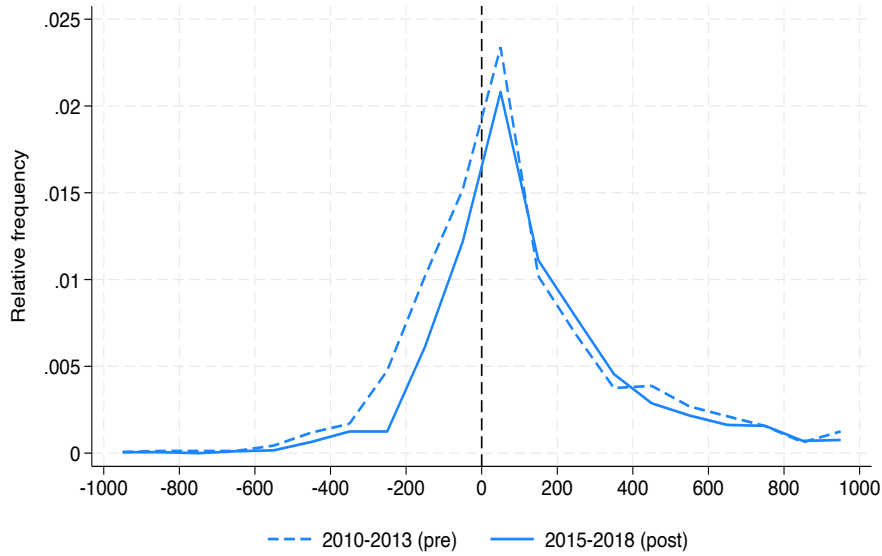


Figure A.7: Spatial distribution of minimum lot size values in 2014

Notes: The figure plots the average minimum lot size value in areas covered by such requirements as of early 2014 in our sample. Light grey areas are zones in our sample that did not use minimum lot sizes requirements. White areas are outside the scope of our study (non-urban areas, missing data, and the Paris municipality). Department limits are represented by thin black lines.



(a) Frequency of minimum lot size values, 2010 to 2013



(b) Density of deviation from the minimum lot size, before and after the reforms

Figure A.8: Observed minimum lot size before and after the 2014 reform

Notes: This figure shows the distribution of observed lot sizes for individual building permits around the 2014 reform. Lot size is measured in square meters. Panel a displays the raw distribution of observed lot sizes in the pre-reform period (2010–2013), distinguishing whether a minimum lot size requirement applied. Panel b plots the deviation between observed and regulatory minimum lot size in the pre- and post-reform periods (2010–2013 and 2015–2018) in areas where minimum lot size applied. Negative values in Panel b indicate that the project fell below the regulatory minimum lot size, while positive values indicate that it exceeded the threshold. Negative deviations may reflect measurement errors (e.g. due to the permits being accepted on the basis of multiple parcels), rounding conventions in lot subdivision, or exceptional permit derogations.

A.4 Additional tables

A.4.1 Real estate outcomes

Table A.4: Main estimations on real estate outcomes

	Log floor space density	Log price per m^2
	(1)	(2)
FAR $\times$ Post	-0.02*	0.01
SE	(0.01)	(0.01)
CI 95%	[-0.05 ; 0.00]	[-0.01 ; 0.02]
R^2	0.14	0.52
N	112,928	564,892
Hedonics		Y

Notes: Columns 1 reports results from the estimation of Equation 2 with log floor space density at the zoning block level as the dependent variable, for the years 2009–2021 (2010 is missing in the density data). Column 2 reports results from the estimation of Equation 1 with log price per square meter as the dependent variable, for the years 2010–2021. The reference year is 2011 in both cases. Price regressions include hedonic controls for single-family dwellings, floor space, number of rooms, and construction vintage. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.5 Robustness checks

A.5.1 Alternative fixed effects specification

Table A.5: TWFE estimates of regulatory responses, alternative FE specification

	Max. height (value) (1)	BCR (value) (2)	Denial probability (3)
FAR $\times$ Post	-0.711***	-0.026***	0.031***
SE	(0.174)	(0.006)	(0.007)
CI 95%	[-1.053 ; -0.370]	[-0.037 ; -0.016]	[0.016 ; 0.045]
R^2	0.50	0.50	0.05
N	4,857,093	4,298,907	87,115

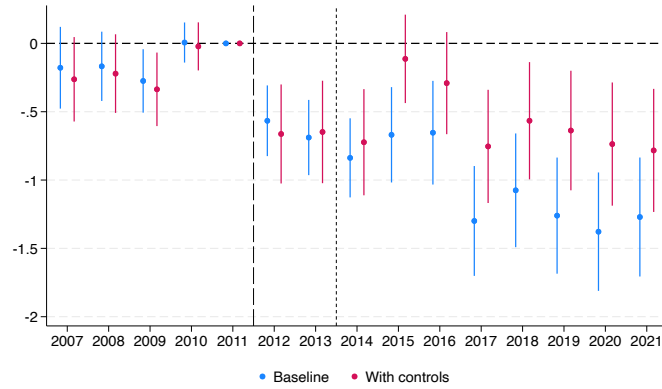
Notes: Results from the estimation of Equation $Y_j = \beta \cdot FAR_{b(j)} \times 1\{t(j) > 2011\} + \alpha_{m(j)} \times FAR_b + \delta_{t(j)} + \epsilon_j$. Column 1 and 2 report linear regressions where the dependent variable is the regulatory value at the parcel level: maximum heights (in meters, column 1) and BCR (ratio of building footprint to parcel size, between 0 and 1, column 2). Column 3 reports a linear probability model where the dependent variable is an indicator equal to 1 if the building permit was denied.. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.6: TWFE estimates on real estate outcomes, alternative FE specification

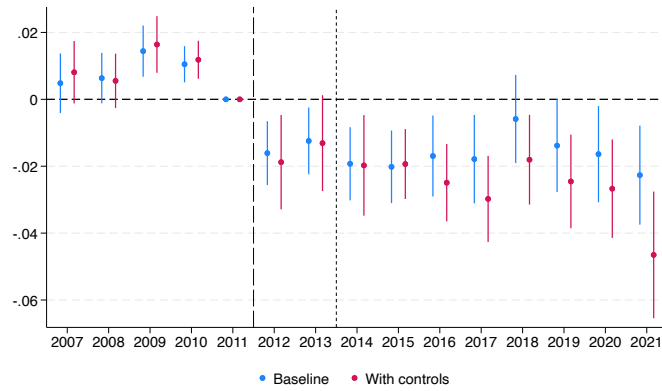
	Log floor space density (1)	Log price per m^2 (2)
FAR $\times$ Post	-0.02*	0.00
SE	(0.01)	(0.01)
CI 95%	[-0.05 ; 0.00]	[-0.01 ; 0.01]
R^2	0.2	0.65
N	112,928	564,892
Hedonics		Y

Notes: Columns 1 reports results from the estimation of Equation $Y_{b,t} = \alpha_{m(b)} \times FAR_b + \delta_t + \beta \cdot FAR_b \times 1\{t > 2011\} + \epsilon_{b,t}$ with log floor space density at the zoning block level as the dependent variable, for the years 2009–2021 (2010 is missing in the density data). Column 2 reports results from the estimation of Equation $Y_j = \beta \cdot FAR_{b(j)} \times 1\{t(j) > 2011\} + \alpha_{m(j)} \times FAR_b + \delta_{t(j)} + \epsilon_j$ with log price per square meter as the dependent variable, for the years 2010–2021. The reference year is 2011. Price regressions include hedonic controls for single-family dwellings, floor space, number of rooms, and construction vintage. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

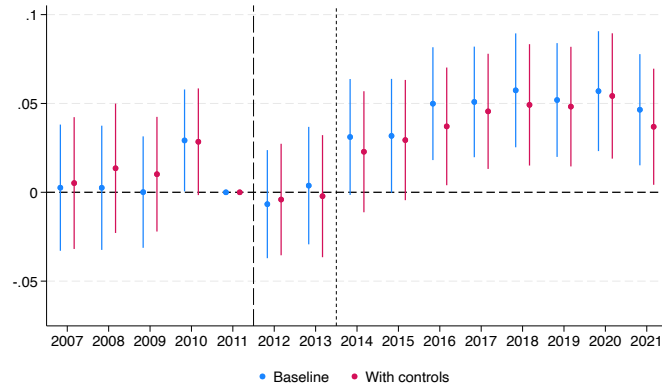
A.5.2 Controls



(a) Maximum heights value



(b) Building coverage ratio value



(c) Building permit denial probability

Figure A.9: Event study for land use regulations with and without controls,
Notes: Results from the estimation of Equation 4 (event study) for the baseline specification without controls (blue) and the baseline with controls (red). Panels (a) and (b) use event study specifications with land use regulation values at the parcel level as the dependent variable. Maximum heights are expressed in meters. BCR values are expressed as ratios of building footprint to parcel size (ranging from 0 to 1). Panel (c) presents a linear probability model where the dependent variable equals 1 if the building permit was denied. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. Vertical bars represent 95% confidence intervals.

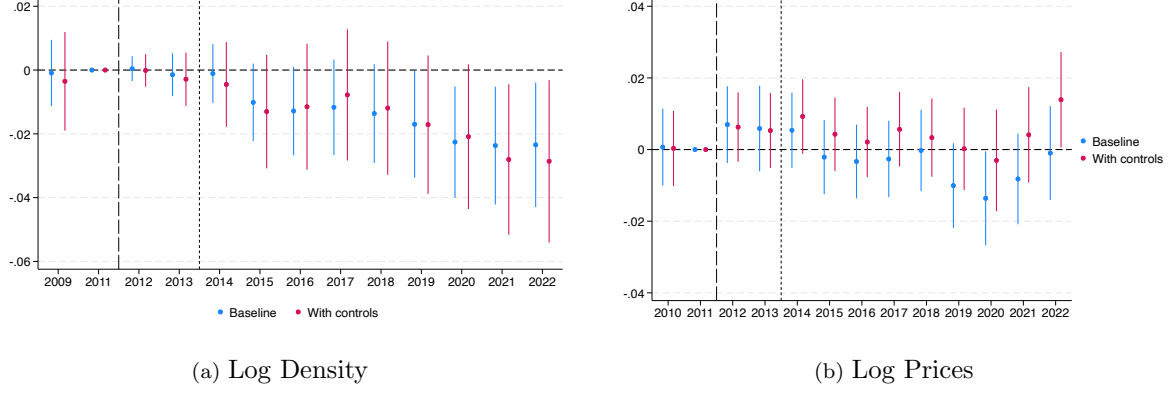


Figure A.10: Event-study estimates for real estate dynamics with and without controls

Notes: Results from the estimation of event-study specifications. Panel (a) plots results using log floor space density as the dependent variable in Equation 4, modified to match the panel structure of the data, as in Equation 2. Panel (b) reports results for log price per square meter as the dependent variable in Equation 1. The event-study specification at the transaction level is $Y_j = \beta_0^{2010} \cdot FAR_{b(j)} \times 1\{t(j) = 2010\} + \sum_{y=2012}^{2021} \beta_1^y \cdot FAR_{b(j)} \times 1\{t(j) = y\} + \alpha_{m(j)} + \delta_{t(j)} + Z_j \cdot \gamma + X_{b(j), 2011} \cdot \eta_{t(j)} + \epsilon_j$, with Z_j a vector of hedonic controls including an indicator for single-family dwellings, floor space, number of rooms, and construction vintage. The reference year is 2011. Regressions include municipal and year fixed effects. Standard errors are clustered at the zoning-block level. Vertical bars represent 95% confidence intervals.

A.5.3 Distances

Table A.7: Robustness to different frontier distances

	(1)	(2)	(3)	(4)
<i>Panel A. Maximum height (value)</i>				
Coeff	-0.82***	-0.82***	-1.10***	-1.28***
SE	(0.18)	(0.19)	(0.21)	(0.22)
95% CI	[-1.16; -0.47]	[-1.20; -0.46]	[-1.52; -0.69]	[-1.71; -0.85]
R^2	0.45	0.44	0.39	0.38
N	4,857,093	3,973,301	3,060,101	1,703,408
<i>Panel B. BCR (value)</i>				
Coeff	-0.02***	-0.01	-0.01**	-0.03***
SE	(0.01)	(0.01)	(0.01)	(0.01)
95% CI	[-0.03; -0.01]	[-0.02; 0.01]	[-0.03; 0.00]	[-0.04; -0.01]
R^2	0.44	0.44	0.43	0.42
N	4,298,907	3,758,525	2,942,851	1,659,188
<i>Panel C. Denial probability</i>				
Coeff	3.05***	3.68***	3.55***	2.38*
SE	(0.72)	(0.84)	(0.97)	(1.23)
95% CI	[1.64; 4.46]	[2.04; 5.32]	[1.64; 5.45]	[-0.03; 4.80]
R^2	0.05	0.05	0.05	0.05
N	86,837	69,118	53,496	30,491
Distance	All	500	250	100

Notes: Results from the estimation of Equation 1. The dependent variable is the regulatory outcome at the parcel or permit level. Panel A reports results for maximum height (in meters), Panel B for BCR (between 0 and 1), and Panel C for denial probability, expressed in percentage points (the denial indicator multiplied by 100). In each panel, column 1 uses the full sample, column 2 restricts to parcels or permits within 500 meters of a treatment frontier, column 3 within 250 meters, and column 4 within 100 meters. All regressions include year fixed effects and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.5.4 Jurisdictional shocks

Table A.8: Robustness specifications on local shocks: land use restrictiveness

	(1)	(2)	(3)
<i>Panel A.</i> Maximum height (value)			
FAR $\times$ Post	-0.82***	-1.09***	-1.95***
SE	(0.18)	(0.20)	(0.35)
95% CI	[-1.16; -0.47]	[-1.49; -0.69]	[-2.63; -1.27]
R^2	0.45	0.45	0.50
N	4,857,093	4,857,086	4,857,086
<i>Panel B.</i> BCR (value)			
FAR $\times$ Post	-0.02***	-0.02***	-0.02*
SE	(0.01)	(0.01)	(0.01)
95% CI	[-0.03; -0.01]	[-0.03; -0.01]	[-0.03; 0.00]
R^2	0.44	0.45	0.56
N	4,298,907	4,298,907	4,298,907
<i>Panel C.</i> Denial probability			
FAR $\times$ Post	0.03***	0.04***	0.03**
SE	(0.01)	(0.01)	(0.01)
95% CI	[0.02; 0.05]	[0.02; 0.05]	[0.00; 0.05]
R^2	0.05	0.05	0.09
N	87,115	87,115	87,083
Time FE	Y	N	N
Municipality FE	Y	Y	Y
Dep. shocks	N	Y	N
Mun. shocks	N	N	Y

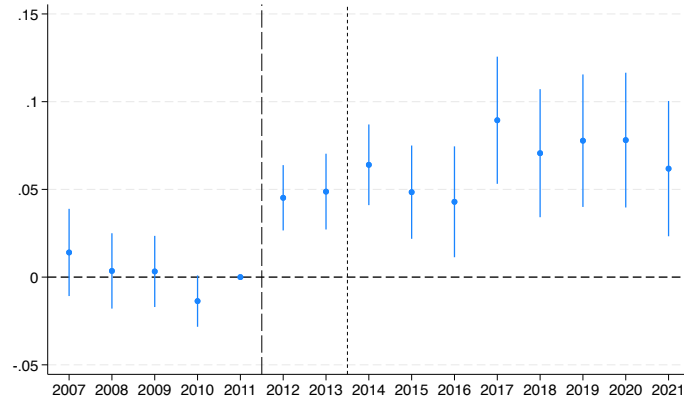
Notes: Results from the estimation of Equation 1. The dependent variable is the regulatory outcome at the parcel or permit level. Panel A reports results for maximum height (in meters), Panel B for BCR (between 0 and 1), and Panel C for denial probability (a probability between 0 and 1). In each panel, column 1 includes time and municipal fixed effects, column 2 includes municipal fixed effects and department-by-year fixed effects, and column 3 includes municipality-by-year fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.9: Robustness specifications on local shocks: real estate dynamics

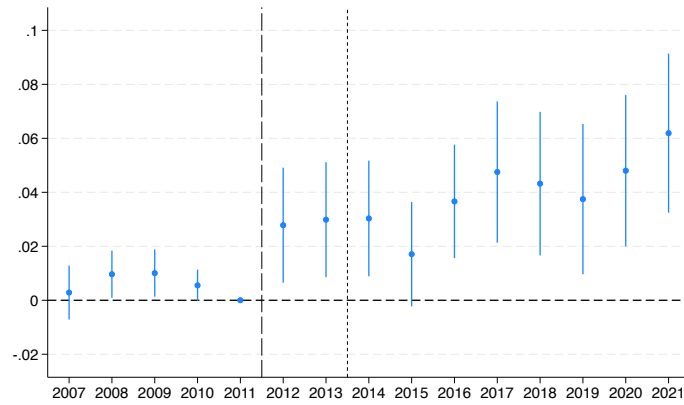
	Log floor space density			Log price per sqm		
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post	-0.02*	-0.02*	-0.03	0.01	0.00	0.01
SE	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
95% CI	[-0.05 ; 0.00]	[-0.05 ; 0.00]	[-0.07 ; 0.01]	[-0.02 ; 0.03]	[-0.02 ; 0.03]	[-0.01 ; 0.04]
R^2	0.14	0.14	0.14	0.52	0.52	0.53
N	112,928	112,928	112,925	564,892	564,892	564,889
Time FE	Y	N	N	Y	N	N
Municipality FE	Y	Y	Y	Y	Y	Y
Dep. shocks	N	Y	N	N	Y	N
Mun. shocks	N	N	Y	N	N	Y

Notes: Columns 1-3 report results from the estimation of Equation 2 with log floor space density at the zoning block level as the dependent variable, for the years 2009–2021 (2010 is missing in the density data). Columns 4-6 report results from the estimation of Equation 1 with log price per square meter as the dependent variable, for the years 2010–2021. Price regressions include hedonic controls for single-family dwellings, floor space, number of rooms, and construction vintage. The reference year is 2011. Regressions in columns 1 and 4 include time and municipal fixed effects. Regressions in columns 2 and 5 include municipal fixed effects and department-by-year fixed effects. Regressions in columns 3 and 6 include municipality-by-year fixed effects. Standard errors are clustered at the zoning block-level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.5.5 Alternative land use regulation measures



(a) Maximum height (dummy $\leq 12m$)



(b) Building coverage ratio (dummy ≤ 0.3)

Figure A.11: Event-study estimates for alternative land use regulation measures

Notes: Results from the estimation of Equation 4. Panels a and b use an event-study specification with the land use regulation dummies at the parcel level as the dependent variable. In Panel a, the dependent variable takes the value 1 if the regulatory maximum height is less than or equal to 12 meters. In Panel b, the dependent variable takes the value 1 if the regulatory BCR is less than or equal to 0.3. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block-level. Vertical bars represent 95% confidence intervals.

Table A.10: Main Regressions for Alternative Land Use Regulation Measures

	Max. height $\leq$ 12m	BCR $\leq$ 0.3
	(1)	(2)
FAR $\times$ Post	0.06***	0.03***
SE	(0.01)	(0.01)
95% CI	[0.03 ; 0.09]	[0.01 ; 0.05]
R^2	0.34	0.36
N	4,857,093	4,298,907

Notes: Results from the estimation of Equation 1 using a linear regression specification with alternative land use regulation measures. The reference year is 2011. The dependent variable for columns 1 and 2 is a dummy taking value 1 if the regulatory maximum height is equal to or lower than 12 meters. The dependent variable for columns 3 and 4 is a dummy taking value 1 if the regulatory BCR is equal to or lower than 0.3. Columns 2 and 4 add control variables measured at baseline and interacted with year dummies at the zoning block-level (log population, log income, age of household head, housing surface per person, and share of social housing). All regressions include municipality fixed effects and year fixed effects. Standard errors are clustered at the zoning block-level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

A.5.6 Political timing

Table A.11: Alternative event study estimations as two shocks

	Max. height (value)	BCR (value)	Denial probability
Anticipation (β_1)	-0.51***	-0.02***	-0.99
SE	(0.16)	(0.01)	(1.05)
95% CI	[-0.83 ; -0.20]	[-0.03 ; -0.01]	[-3.05 ; 1.07]
Post 2014 (β_2)	-0.91***	-0.02***	3.98***
SE	(0.20)	(0.01)	(0.76)
95% CI	[-1.30 ; -0.52]	[-0.04 ; -0.01]	[2.48 ; 5.47]
F-stat ($\beta_1 = \beta_2$)	8.12***	0.63	22.90***
R^2	0.45	0.44	0.05
N	4,857,093	4,298,907	86,837

Notes: Results from the estimation of Equation

$$Y_j = \alpha_{m(j)} + \delta_{t(j)} + \beta_1 \cdot FAR_{b(j)} \times 1_{2012 \leq t(j) \leq 2013} + \beta_2 \cdot FAR_{b(j)} \times 1_{t(j) \geq 2014} + \epsilon_j. \quad (\text{A.1})$$

using a linear regression specification with the land use regulation value at the parcel level as the dependent variable for columns 1 and 2, and the denial probability (expressed in percentage points, i.e. the denial indicator multiplied by 100) for column 3. The reference year is 2011. Values for maximum heights are in meters. Values for BCR are expressed as ratios between building footprint and parcel size (ranging from 0 to 1). The anticipation coefficient, β_1 , corresponds to the treatment effect between the first and the second law, which we label as an “anticipation period”. The Post 2014 coefficient, β_2 , corresponds to the treatment effect after the second law, that effectively removed all FARs. β_2 corresponds to the treatment effect when comparing to the reference period 2011 (not the additional effect). We test for equality between β_1 and β_2 to test whether the second law generated an additional treatment effect by performing a F-test. We report F-stat and p-value from the tests. All regressions include time and municipality fixed effects. Standard errors are clustered at the zoning block-level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

A.6 Heterogeneity

A.6.1 Right-wing and center municipalities

Table A.7.1: Heterogeneity: Right-wing and center-led municipalities in 2008-2020

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-0.97***	-0.02***	-0.01**	0.03***	0.05***
SE	(0.18)	(0.26)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Right (β_3)		2.27***		0.03		-0.03*
SE		(0.54)		(0.02)		(0.01)
FAR $\times$ Right $\times$ Post (β_5)		0.10		-0.02		-0.03*
SE		(0.37)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,857,093	4,298,907	4,298,907	86,764	86,764
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the land use regulation value at the parcel level as the dependent variable. The reference year is 2011. Values for maximum heights are in meters. Values for BCR are expressed as ratios between building footprint and parcel size (ranging from 0 to 1). The common coefficient corresponds to β_1 . The baseline difference between right-wing and left-wing municipalities corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect in areas ruled by right-wing mayors between 2008 and 2020. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block-level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

A.6.2 Victory margins

Table A.7.2: Heterogeneity: Victory margin below median in 2008

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-1.15***	-0.02***	-0.02***	0.03***	0.02**
SE	(0.18)	(0.25)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Margin (β_3)		-1.34**		-0.05***		-0.02
SE		(0.55)		(0.02)		(0.01)
FAR $\times$ Margin $\times$ Post (β_5)		0.63*		-0.00		0.01
SE		(0.34)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.45	0.05	0.05
N	4,857,093	4,857,093	4,298,907	4,298,907	86,764	86,764
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the land use regulation value at the parcel level as the dependent variable. The reference year is 2011. Values for maximum heights are in meters. Values for BCR are expressed as ratios between building footprint and parcel size, ranging from 0 to 1). In this table, $Z_{m(j)}$ is equal to one for municipalities where the mayor's victory margin in the 2008 municipal election is below the sample median. The common coefficient corresponds to β_1 . The baseline relationship between pre-reform FAR constraints and the outcome corresponds to β_2 . The baseline difference in this relationship for municipalities with $Z_{m(j)} = 1$ corresponds to β_3 . The differential post-2011 evolution for municipalities with $Z_{m(j)} = 1$ corresponds to β_4 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect in municipalities where the mayor was elected with a narrow margin in 2008. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block-level. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

A.6.3 Homeownership rate

Table A.7.3: Heterogeneity: Homeownership rate

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-1.05***	-0.02***	-0.01	0.03***	0.04***
SE	(0.18)	(0.33)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Homeowner (β_3)		1.34**		0.01		-0.02
SE		(0.57)		(0.02)		(0.02)
FAR $\times$ Homeowner $\times$ Post (β_5)		0.50		-0.02		-0.02
SE		(0.40)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,857,093	4,298,907	4,298,907	86,764	86,764
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a share of homeowners above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.4 Detached homes

Table A.7.4: Heterogeneity: Detached homes

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-1.18***	-0.02***	-0.02*	0.03***	0.03***
SE	(0.18)	(0.39)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Detached (β_3)		1.52***		0.02		-0.02
SE		(0.58)		(0.02)		(0.01)
FAR $\times$ Detached $\times$ Post (β_5)		0.64		-0.01		-0.01
SE		(0.43)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a share of detached (single family) home above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.5 Share of open land in municipality

Table A.7.5: Heterogeneity: open land

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-0.48**	-0.02***	-0.02***	0.03***	0.05***
SE	(0.18)	(0.22)	(0.01)	(0.00)	(0.01)	(0.01)
FAR $\times$ Open land (β_3)		3.12***		0.03*		0.02
SE		(0.54)		(0.02)		(0.02)
FAR $\times$ Open land $\times$ Post (β_5)		-0.50		-0.00		-0.05***
SE		(0.34)		(0.01)		(0.01)
R^2	0.45	0.46	0.44	0.44	0.05	0.05
N	4,857,093	4,857,093	4,298,907	4,298,907	86,764	86,764
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a share of open space above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.6 Housing Prices

Table A.7.6: Heterogeneity: Housing prices

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-0.41	-0.02***	-0.03***	0.03***	0.01
SE	(0.18)	(0.27)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Price (β_3)		-2.53***		-0.03*		-0.04**
SE		(0.56)		(0.02)		(0.02)
FAR $\times$ Price $\times$ Post (β_5)		-0.78**		0.01		0.04***
SE		(0.36)		(0.01)		(0.01)
R^2	0.45	0.46	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with average housing prices above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.7 Median municipal income

Table A.7.7: Heterogeneity: Municipal income

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-0.59**	-0.02***	-0.01**	0.03***	0.05***
SE	(0.18)	(0.28)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Income (β_3)		2.12***		0.00		-0.03**
SE		(0.55)		(0.02)		(0.01)
FAR $\times$ Income $\times$ Post (β_5)		-0.67*		-0.01		-0.03*
SE		(0.36)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a median municipal income above the sample median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.8 Social housing

Table A.7.8: Heterogeneity: Social Housing

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-1.16***	-0.02***	-0.02**	0.03***	0.01
SE	(0.18)	(0.25)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Social (β_3)		-2.09***		-0.01		0.03*
SE		(0.55)		(0.02)		(0.01)
FAR $\times$ Social $\times$ Post (β_5)		0.34		0.00		0.04***
SE		(0.38)		(0.01)		(0.01)
R^2	0.45	0.45	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a share of social housing above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6.9 Share of people above 65 years old

Table A.7.9: Heterogeneity: share of people above 65

	Max. height (value)		BCR (value)		Denial probability	
	(1)	(2)	(3)	(4)	(5)	(6)
FAR $\times$ Post (β_1)	-0.82***	-1.46***	-0.02***	-0.01**	0.03***	0.05***
SE	(0.18)	(0.33)	(0.01)	(0.01)	(0.01)	(0.01)
FAR $\times$ Above 65 (β_3)		2.60***		0.02		-0.00
SE		(0.57)		(0.02)		(0.01)
FAR $\times$ Above 65 $\times$ Post (β_5)		1.43***		-0.02		-0.04**
SE		(0.40)		(0.01)		(0.01)
R^2	0.45	0.46	0.44	0.44	0.05	0.05
N	4,857,093	4,856,995	4,298,907	4,298,903	86,764	86,747
Time FE	Y	Y	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y	Y	Y
Heterogeneity	N	Y	N	Y	N	Y
Controls	N	N	N	N	N	N

Notes: Results from the estimation of Equation 5 using a linear regression specification with the parcel-level land use regulation outcome as the dependent variable. The reference year is 2011. Maximum height is measured in meters. BCR is defined as the ratio of building footprint to parcel size (0 to 1). Denial probability of submitted permits in columns 5 and 6. The common coefficient corresponds to β_1 . The baseline difference coefficient corresponds to β_3 . The interaction coefficient corresponds to β_5 and measures the additional treatment effect of living in an area with a share of people above 65 above the median. All regressions include year and municipal fixed effects. Standard errors are clustered at the zoning block level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.